



COMPLETE PROJECT REPORT

Build it. Prove it. Defend it.

A privacy-first career preparation system that connects resume parsing, role evidence, interviews, honest feedback and targeted growth.

27 SEPTEMBER 2026 · PROOF COACH AI

PROOF COACH AI — SYSTEM & PRODUCT DEEP-DIVE REPORT

Tagline: Build it. Prove it. Defend it.

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| 1. Executive Overview

1.1 The Core Identity

ProofCoach AI is a privacy-first, local-first career preparation platform designed to replace ungrounded resume buzzwords with machine-readable, evidence-linked, and interview-defensible technical competencies. Its guiding philosophy is captured in its three-part operational standard: 1. **Build it:** Turn theoretical study into tangible software artifacts and code repositories. 2. **Prove it:** Extract measurable claims, ground them in verifiable work, and test layout readability through transparent parser diagnostics. 3. **Defend it:** Withstand targeted, adaptive technical interview probing that interrogates baselines, personal contributions, trade-offs, and failure scenarios.

1.2 The Central Product Promise

Contemporary job candidates face an acute paradox: while Generative AI enables individuals to produce polished, impressive-sounding resumes in seconds, hiring algorithms and technical interviewers increasingly reject applicants who cannot substantiate their claims under pressure. ProofCoach AI operates under a foundational principle:

"A resume should not merely sound impressive. Machines must be able to parse its structure reliably, the candidate must possess verifiable evidence for every quantified claim, and the candidate must be prepared to defend those claims under rigorous technical interrogation."

1.3 Key Architectural Pillars

- **Local-First & Privacy-Governed:** The platform runs locally via FastAPI and SQLite, with client-side execution in React 19. It enforces a strict zero-telemetry policy, executes document parsing in RAM, and provides a one-click local data destruction cascade.
- **Evidence Lock Safeguard:** Prevents generative hallucination by actively blocking any resume rewrite that introduces unsupported metrics, unverified technologies, or fabricated impact figures.
- **Transparent Multi-Metric Diagnostic:** Completely rejects the concept of a single, opaque "AI Readiness Score" or "Universal ATS Score." Instead, it surfaces six independent, observable signals: Parser Robustness, Role Evidence Coverage, Interview Readiness, Claim Verification, Learning Progress, and Career Quest XP.
- **Deterministic Vertical Slice:** All core features operate deterministically out of the box without requiring external cloud AI subscriptions, API keys, or cloud infrastructure, while maintaining an optional abstraction layer ([AIProvider](#)) for local

2. What Is ProofCoach AI?

2.1 The Genesis of the Platform

ProofCoach AI was engineered to dismantle the superficiality of modern career preparation. Traditional tools fall into three segregated silos: 1. **Resume Optimizers**: Focus exclusively on keyword stuffing and stylistic rephrasing, often hallucinating achievements that candidates cannot defend. 2. **Mock Interview Simulators**: Ask generic, disconnected behavioral questions (e.g., "Tell me about a time you failed") without contextual grounding in the candidate's actual projects. 3. **Online Learning Platforms**: Prescribe generic, open-ended course catalogs without diagnosing the candidate's exact technical gaps or requiring concrete artifact construction.

ProofCoach AI bridges these silos into a single closed-loop ecosystem. A claim extracted from a resume directly triggers targeted interview questions; weaknesses exposed during the interview immediately generate precise learning goals; learning goals culminate in tangible project tasks in the Project Lab; and finished project artifacts loop back to strengthen the resume under the supervision of Evidence Lock.

2.2 Target User Personas

The system explicitly caters to four distinct user tiers:

CANDIDATE CATEGORY	CORE CHALLENGE ADDRESSED	PROOFCOACH VALUE DRIVER
Student	Academic projects lack commercial context and quantified metrics.	Translates coursework into defensible artifacts; extracts baseline measurements.
Fresher / New Grad	Limited professional history leads to weak keyword matching.	Builds verifiable project evidence to bypass ATS parsing hurdles; closes role gaps.
Early-Career Professional	Difficulty differentiating personal contributions from team work.	Pressure-tests ownership, architectural trade-offs, and optimization benchmarks.
Career Switcher	Existing experience does not directly align with new role titles.	Maps transferable skills to target requirements; builds focused bridge projects.

3. Product Philosophy & Responsible-AI Boundaries

ProofCoach AI adheres to strict ethical and operational boundaries that separate verifiable reality from simulated or fabricated assertions.

3.1 Anti-Hallucination Policy (Evidence Lock)

Generative AI often produces persuasive lies. If an applicant writes **"Made API faster"**, common AI rewrites suggest **"Reduced API response latency by 45% using Redis caching"**. If the candidate never used Redis and never measured latency, they will fail the technical screening. Evidence Lock acts as an immutable gatekeeper: - **Green (Safe)**: Structural, grammatical, or clarity enhancements with zero new factual claims. - **Amber (Confirm)**: Semantic changes that elevate ownership (e.g., from **"participated"** to **"architected"**), requiring explicit user confirmation. - **Red (Blocked)**: Introduction of unconfirmed percentages, numbers, tools, or certifications. The system outright blocks these edits.

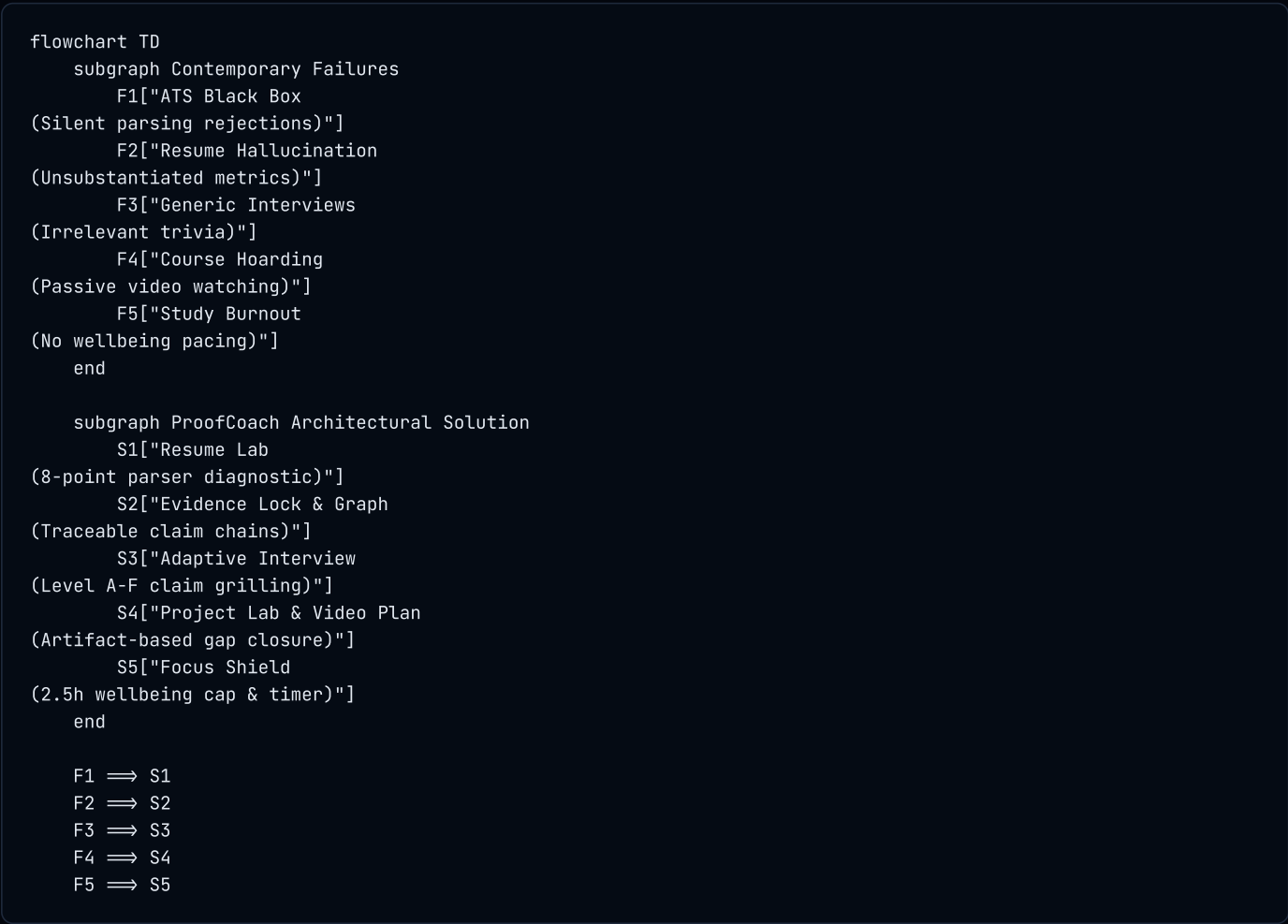
3.2 Observable Communication Indicators vs. Fake Emotional Analysis

ProofCoach AI vehemently rejects the pseudo-scientific practice of facial emotion analysis, voice stress analysis, or visual "confidence scoring" through webcams: - **What is measured:** Objective, observable signals—word count, speech duration, filler-word frequency (`um` , `uh` , `like`), answer structure (Problem-Approach-Measurement-Tradeoff), and topic relevance. - **What is avoided:** Facial expression tracking, eye-gaze tracking, micro-expression decoding, and AI-inferred emotional traits.

3.3 Voluntary Gender & Non-Inference

The system strictly prohibits inferring candidate demographic characteristics. Gender, ethnicity, and personal background are never guessed from names, resumes, or vocal traits. A voluntary gender field is provided exclusively for candidate-driven preferences (e.g., auto-pairing with a preferred AI voice persona) and remains hidden unless explicitly configured.

4. Problem & Solution: The Crisis in Career Prep



The Five Core Structural Breakdowns:

- The ATS Black Box:** Traditional candidates submit multi-column or heavily formatted resumes that Applicant Tracking Systems scramble or drop entirely. Candidates receive no actionable feedback on why their document failed extraction.
- ProofCoach Fix:** An 8-point local diagnostic parser that exposes exactly how machines parse the text, highlighting column ambiguities, missing headers, and unparsed tables.
- Resume Inflation:** Candidates copy AI-generated bullet points featuring grandiose claims (e.g., "Scaled database to 1M QPS"). When asked for p99 latency figures in interviews, they freeze.

4. **ProofCoach Fix:** Claim extraction algorithms identify every quantified statement and force the candidate to verify it in the Evidence Graph before locking it into the resume.
 5. **Disconnected Mock Interviews:** Interview platforms ask generic behavioral or leetcode questions completely decoupled from what is written on the candidate's CV.
 6. **ProofCoach Fix:** The Adaptive Interview engine extracts the candidate's specific claims (e.g., "Reduced latency by 35%") and initiates an evidence challenge specifically probing that sentence.
 7. **Passive Learning:** Students spend hundreds of hours watching tutorials without writing code, retaining very little practical knowledge.
 8. **ProofCoach Fix:** The Video Study Planner and Project Lab enforce an active 75% learning / 25% practice split, requiring learners to build concrete code artifacts for every studied module.
 9. **Burnout and Overwork:** High-stress job preparation often results in endless, unstructured cramming sessions with diminishing cognitive returns.
 10. **ProofCoach Fix:** The Focus Shield establishes strict 150-minute study limits, provides session autosaving, and suppresses in-app noise to encourage healthy learning habits.
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| 5. Complete Product Map

ProofCoach AI comprises **13 distinct client routes** organized into four functional quadrants:

flowchart TD

subgraph ProofCoach Workspace

direction TB

subgraph Onboarding & Identity

R1["/ (Landing Page)"]

R2["/onboarding (Candidate Setup)"]

R3["/profile (Command Centre)"]

R4["/dashboard (Legacy Redirect)"]

end

subgraph Evidence & Alignment

R5["/resume (Resume Lab)"]

R6["/role (Target Role)"]

R7["/evidence (Evidence Graph)"]

end

subgraph Verification & Challenge

R8["/interview (Adaptive Interview)"]

R9["/feedback (Evidence Feedback)"]

end

subgraph Growth Loop & Execution

R10["/learn (Learning Plan)"]

R11["/video-plan (Video Study Planner)"]

R12["/project-lab (Project Lab)"]

R13["/quest (Career Quest)"]

R14["/focus (Focus Shield)"]

end

subgraph System Governance

R15["/settings (Privacy & Runtime)"]

end

end

R1 → R2 → R3

R3 → R5 → R6 → R7 → R8 → R9 → R10

R10 → R11

R10 → R12

R12 → R13

R10 → R14

R3 -.-> R15

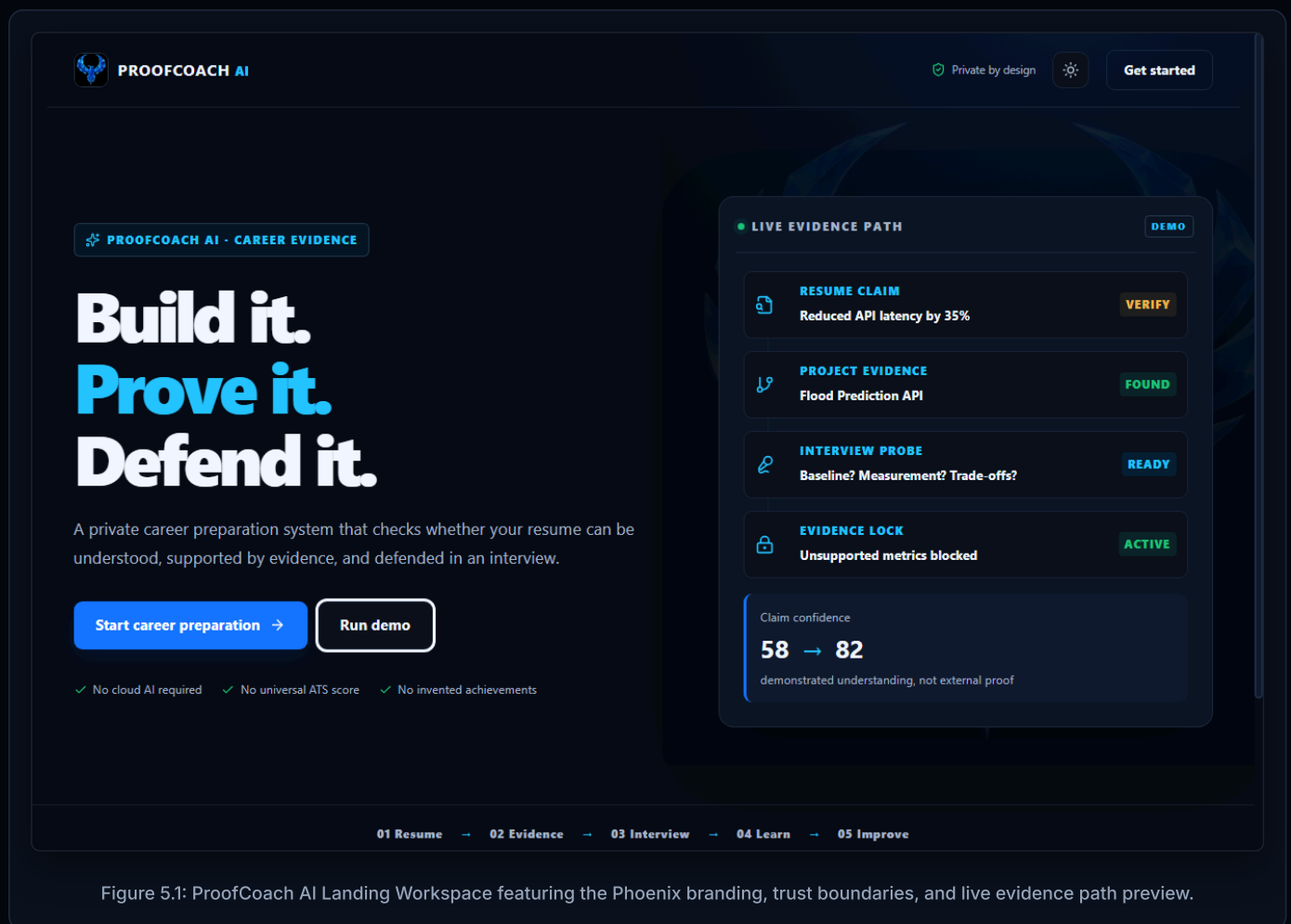


Figure 5.1: ProofCoach AI Landing Workspace featuring the Phoenix branding, trust boundaries, and live evidence path preview.

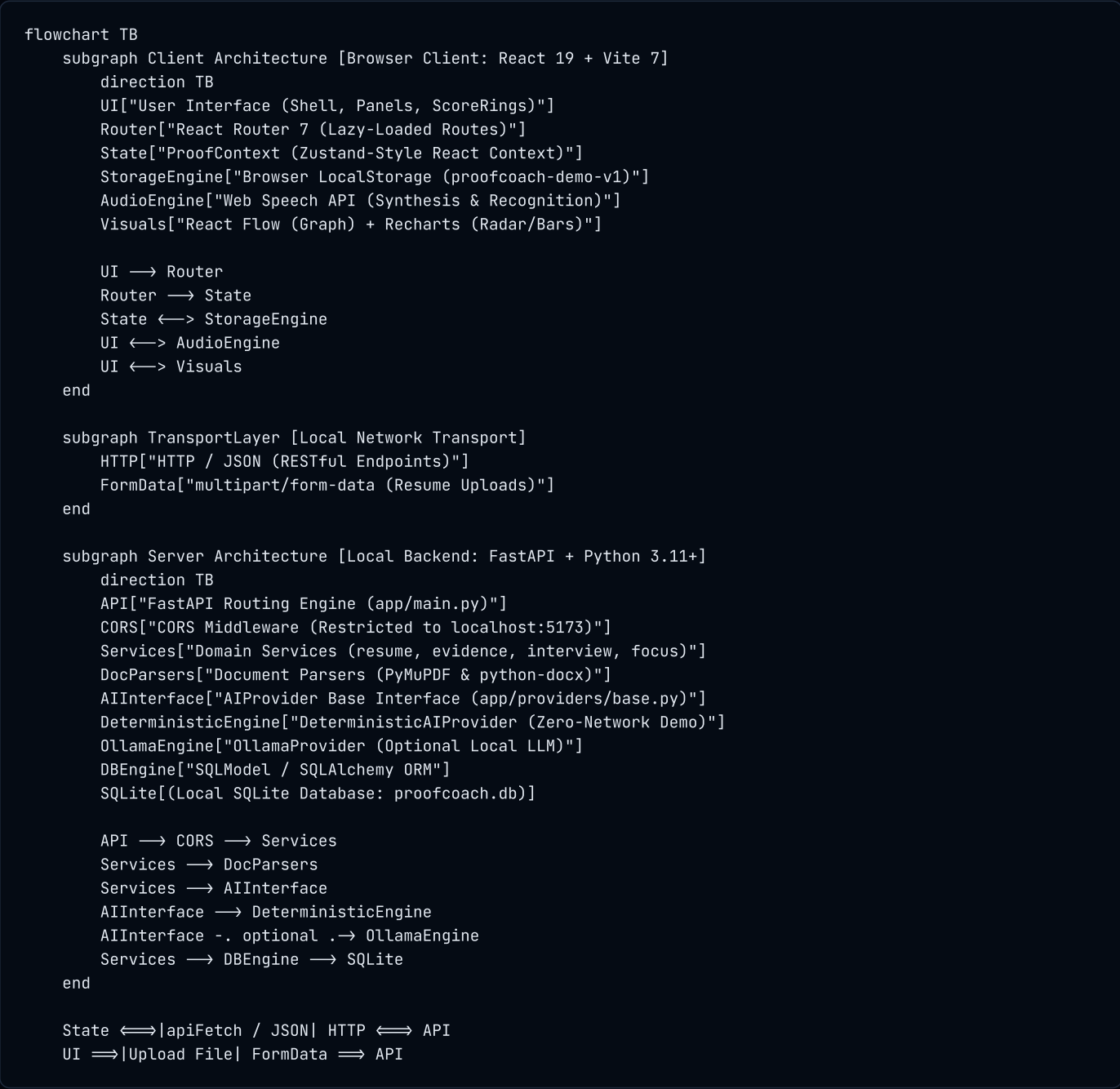
Route Index and Component Responsibilities

ROUTE	VIEW COMPONENT	CORE PURPOSE & USER CAPABILITIES	DATA DEPENDENCIES
<code>/</code>	<code>LandingPage.tsx</code>	High-impact product overview, feature cards, deterministic demo entry point.	Static demo fixtures
<code>/onboarding</code>	<code>OnboardingPage.tsx</code>	3-step setup: Candidate details, target role selection, practice/interviewer preferences.	<code>ProofContext</code>
<code>/profile</code>	<code>ProfilePage.tsx</code>	Comprehensive candidate dossier embedding the central command center metrics.	<code>DemoState.candidate</code> , <code>metrics</code>
<code>/dashboard</code>	<code>DashboardPage.tsx</code>	Redirects to <code>/profile</code> ; provides the 6-score diagnostic grid, priority alerts, and next 48h tasks.	<code>DemoState.metrics</code> , <code>completed</code>
<code>/resume</code>	<code>ResumePage.tsx</code>	Drag-and-drop PDF/DOCX parser, 8-part robustness breakdown, measurable claim detector.	Local FastAPI <code>/api/resume/parse</code>
<code>/role</code>	<code>RolePage.tsx</code>	Role selector & custom job description parser; skill coverage matrix & gap extraction.	Local FastAPI <code>/api/role/analyze</code>
<code>/evidence</code>	<code>EvidencePage.tsx</code>	Interactive React Flow DAG mapping candidate skills to projects, claims, and role specs.	<code>DemoState.candidate</code> , local API
<code>/interview</code>	<code>InterviewPage.tsx</code>	Real-time adaptive interview arena; Web Speech audio synthesis & transcription; level A-F probes.	Local FastAPI <code>/api/interview/*</code>
<code>/feedback</code>	<code>FeedbackPage.tsx</code>	Recharts radar breakdown, claim truth loop (before vs after), Evidence Lock rewrite sandbox.	<code>DemoState.evaluation</code> , local API
<code>/learn</code>	<code>LearnPage.tsx</code>	Prioritized gap-closure curriculum; verified free documentation links; daily study schedules.	Local FastAPI <code>/api/learning/plan</code>
<code>/video-plan</code>	<code>VideoPlanPage.tsx</code>	YouTube embed study orchestrator; timestamped note capture; 5-tier comprehension funnel.	<code>DemoState.video</code> , <code>video.ts</code>
<code>/project-lab</code>	<code>ProjectLabPage.tsx</code>	9-stage engineering journey; Kanban task organizer; resume evidence generation.	<code>DemoState.project</code>
<code>/quest</code>	<code>QuestPage.tsx</code>	Gamified progression system; 3 tiers (Build, Prove, Perform); transparent XP reward rules.	<code>DemoState.metrics.questPoints</code>
<code>/focus</code>	<code>FocusPage.tsx</code>	Distraction-reduced workspace; 150m wellbeing countdown; 20s demo timer; autosave trigger.	<code>localStorage</code> (<code>proofcoach-focus-v1</code>)

ROUTE	VIEW COMPONENT	CORE PURPOSE & USER CAPABILITIES	DATA DEPENDENCIES
<code>/settings</code>	<code>SettingsPage.tsx</code>	Dark/light theme switch, runtime mode inspector, JSON progress export, local data purge.	<code>ProofContext</code> , <code>/api/data</code>

6. System Architecture

ProofCoach AI employs a modern, decoupled client-server architecture engineered specifically for local execution, low resource overhead, and strict privacy boundaries.



6.1 Architectural Subsystems

- Presentation Layer (React 19 SPA):** Utilizes Vite for bundling. Reusable atomic UI primitives (`Panel` , `ScoreRing` , `StatusPill` , `PageIntro`) ensure visual consistency across both Dark Navy and White Editorial themes.

2. **State & Reactive Store** (`ProofContext.tsx`): Maintains application-wide reactivity. Persists non-sensitive candidate configuration, active project state, video notes, and diagnostic scores to `localStorage` under key `proofcoach-demo-v1`.
 3. **Transport Layer** (`api.ts`): Centralizes all HTTP communication with the backend (`http://localhost:8000`). Automatically handles network timeouts, JSON parsing, error marshalling, and transparent fallback to local demo data if the server is unreachable.
 4. **Backend Application** (`app/main.py`): Built with FastAPI and Starlette. Implements asynchronous lifespan management to automatically initialize the SQLite database schema on startup without external migration scripts.
 5. **Document Extraction Subsystem** (`services/resume.py`): Uses PyMuPDF (`fitz`) for PDF page extraction, character counting, and bounding-box spatial analysis to identify multi-column formatting. Uses `python-docx` to traverse paragraph and table XML trees.
 6. **Heuristic & Rule Engines**: Pure Python service functions execute claim detection, skill extraction, coverage scoring, answer evaluation, and Evidence Lock filtering using deterministic pattern matching.
 7. **Persistence Subsystem** (`app/db.py` & `app/models.py`): Built with SQLAlchemy. Manages local tables for `CandidateProfile`, `ResumeRecord`, and `InterviewSession` within an embedded SQLite database.
-

| 7. Complete User Journey: A Day in the Life

To understand how ProofCoach operates in practice, consider the end-to-end journey of **Alex**, a second-year AI & Machine Learning student preparing for a Backend Developer role.

```
sequenceDiagram
    autonumber
    actor User as Alex (Candidate)
    participant UI as ProofCoach UI
    participant Ctx as ProofContext (Browser)
    participant API as FastAPI Backend
    participant Doc as PyMuPDF / docx
    participant DB as SQLite DB

    Note over User, UI: Phase 1: Onboarding & Setup
    User->>UI: Enters Name, Education, Target Role (Backend Dev)
    UI->>Ctx: updateCandidate() → commits to localStorage
    UI->>User: Displays Command Centre with initial baseline metrics

    Note over User, UI: Phase 2: Resume Lab & Parsing
    User->>UI: Uploads resume.pdf (contains 35% latency claim)
    UI->>API: POST /api/resume/parse (multipart form)
    API->>Doc: Extract text blocks & layout coordinates
    Doc-->>API: Returns text + detected two-column layout
    API->>API: Calculate Parser Robustness (82/100) & Extract claims
    API-->>UI: Returns ParserAnalysis JSON
    UI->>User: Displays layout warnings & detected claims

    Note over User, UI: Phase 3: Alignment & Evidence Mapping
    User->>UI: Selects Target Role (Backend Developer)
    UI->>API: POST /api/role/analyze
    API-->>UI: Required: Python, REST, SQL, Docker, Testing, SysDesign
    UI->>Ctx: Matches candidate skills → identifies 3 gaps (Docker, Testing, SysDesign)
    User->>UI: Inspects Career Evidence Graph
    UI->>User: Renders interactive DAG showing broken link at Docker

    Note over User, UI: Phase 4: Live Adaptive Interview
    User->>UI: Launches Interview for "Reduced latency by 35%"
    UI->>API: POST /api/interview/start
    API->>DB: Create InterviewSession record
    API-->>UI: Returns Question 1 (Level D: Evidence Verification)
    UI->>User: Speaks question aloud via Web Speech API
    User->>UI: Speaks/Types answer ("Baseline was 420ms, used caching...")
    User->>UI: Clicks "Submit evidence"
    UI->>API: POST /api/interview/{id}/answer
    API->>API: evaluate_answer() → scores dimensions & structure
    API->>DB: Append transcript JSON
    API-->>UI: Returns evaluation & Next Question (Level E: Trade-offs)
    UI->>Ctx: submitInterview(evaluation) → updates XP (+50 XP)

    Note over User, UI: Phase 5: Feedback & Evidence Lock
    UI->>User: Shows radar chart, truth loop (+24% claim confidence)
    User->>UI: Tests rewrite: "Reduced API latency by 40%"
    UI->>API: POST /api/evidence-lock/rewrite
    API-->>UI: Status BLOCKED (40% is unsupported)
    UI->>User: Shows Red Warning: Evidence Lock blocked fabrication

    Note over User, UI: Phase 6: Targeted Gap Closure
    User->>UI: Opens Personalized Learning Plan
    UI->>API: POST /api/learning/plan (gaps: Docker, Testing)
    API-->>UI: Returns structured curriculum & verified doc links
    User->>UI: Opens Project Lab → creates "FastAPI Containerization"
    User->>UI: Completes project task → progress updates to 24%
    User->>UI: Activates Focus Shield → 150m timer counts down
    UI->>Ctx: Autosaves session state locally
```

8. Profile & Command Centre

8.1 Overview & Purpose

The Profile & Command Centre (`/profile`) serves as the central administrative cockpit of ProofCoach AI. It brings together the candidate's personal identity, academic background, target role parameters, daily practice constraints, and interviewer configuration alongside real-time diagnostic performance metrics.

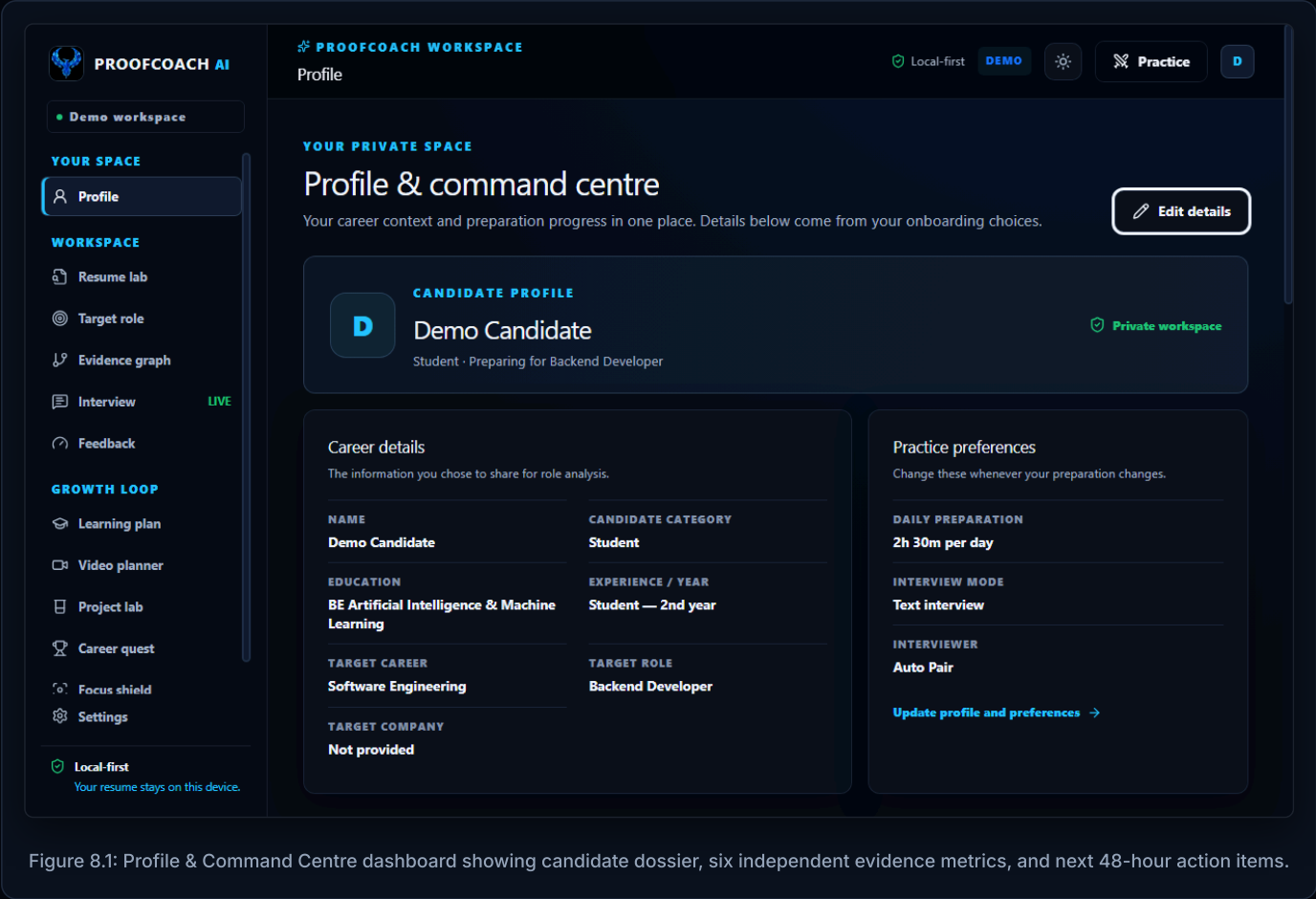


Figure 8.1: Profile & Command Centre dashboard showing candidate dossier, six independent evidence metrics, and next 48-hour action items.

8.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The candidate sees their name, educational status, target career, and daily study goal. Directly below, six diagnostic scores summarize how well their resume parses, how many skills are backed by evidence, how ready they are for interviews, how well their claims hold up, how much learning they have completed, and their current Career Quest rank. A "Next 48 Hours" action panel highlights the single most important task to work on today.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

When `/profile` mounts, the component extracts the `candidate` object and `metrics` record from `ProofContext`. The daily time budget (stored as total minutes, e.g., 150) is formatted into human-readable hours and minutes (`2h 30m per day`). If the candidate has configured voluntary gender and selected the "Auto Pair" persona, the system renders a voluntary badge; otherwise, demographic fields remain entirely hidden. The embedded `DashboardPage` calculates the overall readiness distribution across six distinct metrics without ever combining them into a single black-box score.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/ProfilePage.tsx` , `frontend/src/pages/DashboardPage.tsx`
- **Component Hierarchy:** `Shell` → `ProfilePage` → `Panel` , `Detail` , `DashboardPage` → `ScoreRing` , `StatusPill`
- **State Consumed:** `state.candidate: Candidate` , `state.metrics: Record<MetricKey, number>` , `state.completed: string[]`

- **API Dependencies:** None for initial display; data is served directly from the reactive `ProofContext` store initialized from `localStorage`.
- **Persistence:** State updates trigger `localStorage.setItem('proofcoach-demo-v1', JSON.stringify(next))`.
- **Failure Modes:** If `localStorage` is empty or corrupted, `loadState()` gracefully catches the JSON parse error and populates the view with `initialDemoState`.

9. Resume Lab & Parser Robustness Engine

9.1 Overview & Purpose

The Resume Lab (`/resume`) provides an unvarnished, transparent diagnostic of how automated applicant-tracking systems and machine parsers interpret an uploaded resume. It explicitly rejects commercial "ATS Score" claims, instead computing a multi-attribute **Parser Robustness Score (0–100)** with spatial layout warnings.

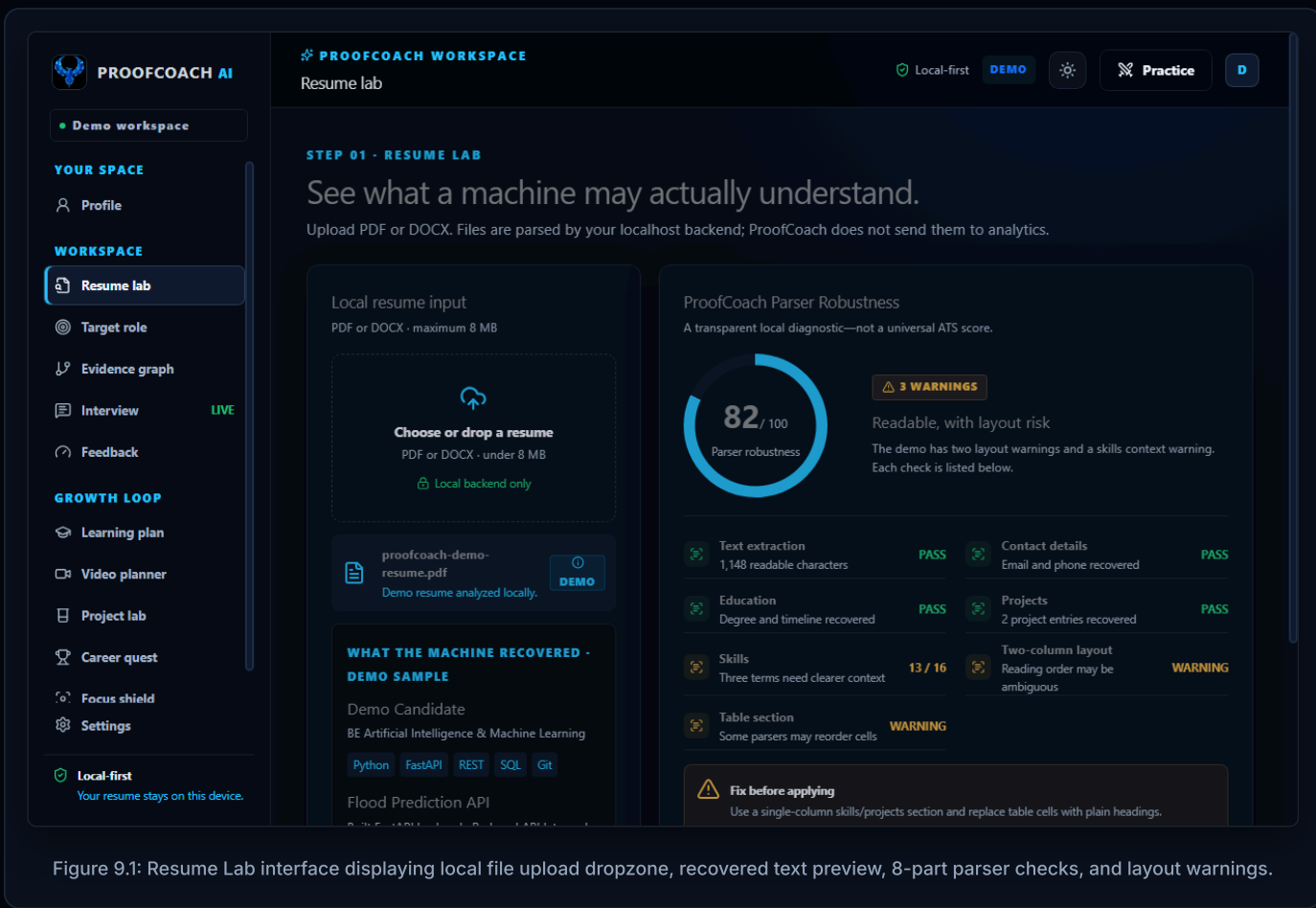


Figure 9.1: Resume Lab interface displaying local file upload dropzone, recovered text preview, 8-part parser checks, and layout warnings.

9.2 The Eight-Point Robustness Rubric

The local parser evaluates documents across 8 distinct criteria totaling 100 points:

CHECK ITEM	MAXIMUM POINTS	PASSING CONDITION	WARNING CONDITION & DIAGNOSTIC
1. Text Extraction	20	Text ratio ≥ 0.75 (≥ 675 chars)	Less than 675 chars recovered; points scaled proportionally.
2. Contact Details	10	Valid email AND phone regex match	Missing email or phone format; awards 5/10.
3. Recognizable Sections	15	≥ 3 standard section headers found	Fewer than 3 headers; awards 3 pts per detected header.
4. Timeline Dates	10	Year (19xx/20xx) or Month names found	No recognized timeline markers; awards 4/10.
5. Skill Vocabulary	15	≥ 4 recognized catalog skills found	Fewer than 4 skills; awards 3 pts per detected skill.
6. Projects Section	10	Project section header recovered	Project header missing; awards 0/10.
7. Two-Column Layout	10	Single column reading order	Two-column layout detected via horizontal block coordinates; penalizes 6 pts (awards 4/10).
8. Table Content	10	No tables detected (linear text)	Word tables detected; penalizes 6 pts (awards 4/10).

9.3 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The user drags and drops a PDF or DOCX file (under 8 MB). The screen immediately shows what text the machine was able to extract, highlights any detected skills, and lists every measurable claim found in the document. A large score ring displays the Parser Robustness score with clear warnings (e.g., "Two-column layout warning: reading order may be ambiguous").

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

- In local mode, the file is transmitted as a `multipart/form-data` payload to `POST /api/resume/parse`.
- PyMuPDF (`fitz`) opens the PDF stream in RAM. It computes the midpoint of each page (`rect.width / 2`). If bounding boxes are found on both the left and right of the midpoint simultaneously, `layout["two_column"]` is set to `True`.
- For DOCX files, `python-docx` traverses paragraphs and table cell text, setting `layout["tables"] = True` if any table objects exist.
- Regex filters scan for measurable claims using a pattern matching percentages, numerical multipliers, and active impact verbs.
- The 8 checks are computed, summed, and returned alongside an 800-character plain-text preview.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- Source Files:** `frontend/src/pages/ResumePage.tsx`, `backend/app/services/resume.py`, `backend/app/main.py`
- Functions:** `parseResume(file: File)`, `extract_text()`, `analyze_resume_text()`, `extract_claims()`
- Regex Patterns:**
- Claims:** `([\d+](?:\.\d+)?\s*%|[\d+][kKmM]?[+]?[s+](?:users?|requests?|clients?))| (?:reduced|increased|improved|accelerated|saved|grew|led|managed|scaled|optimized))`

- Email: `[\\w.+-]+@[\\w.-]+\\. [A-Za-z]{2,}`
- Phone: `(?:\\+?\\d{\\s-}?) {8,14}`
- **Fallback Behavior:** If the FastAPI backend is offline, `ResumePage` catches the network error, sets `status` to a transparent warning, and renders the deterministic demo analysis (82/100 robustness, 2 layout warnings).

10. Target Role Analysis

10.1 Overview & Purpose

The Target Role module (`/role`) translates ambiguous job postings into clear evidence requirements. Rather than performing simple keyword density counts, it categorizes skills into required competencies, preferred tools, and core engineering responsibilities.

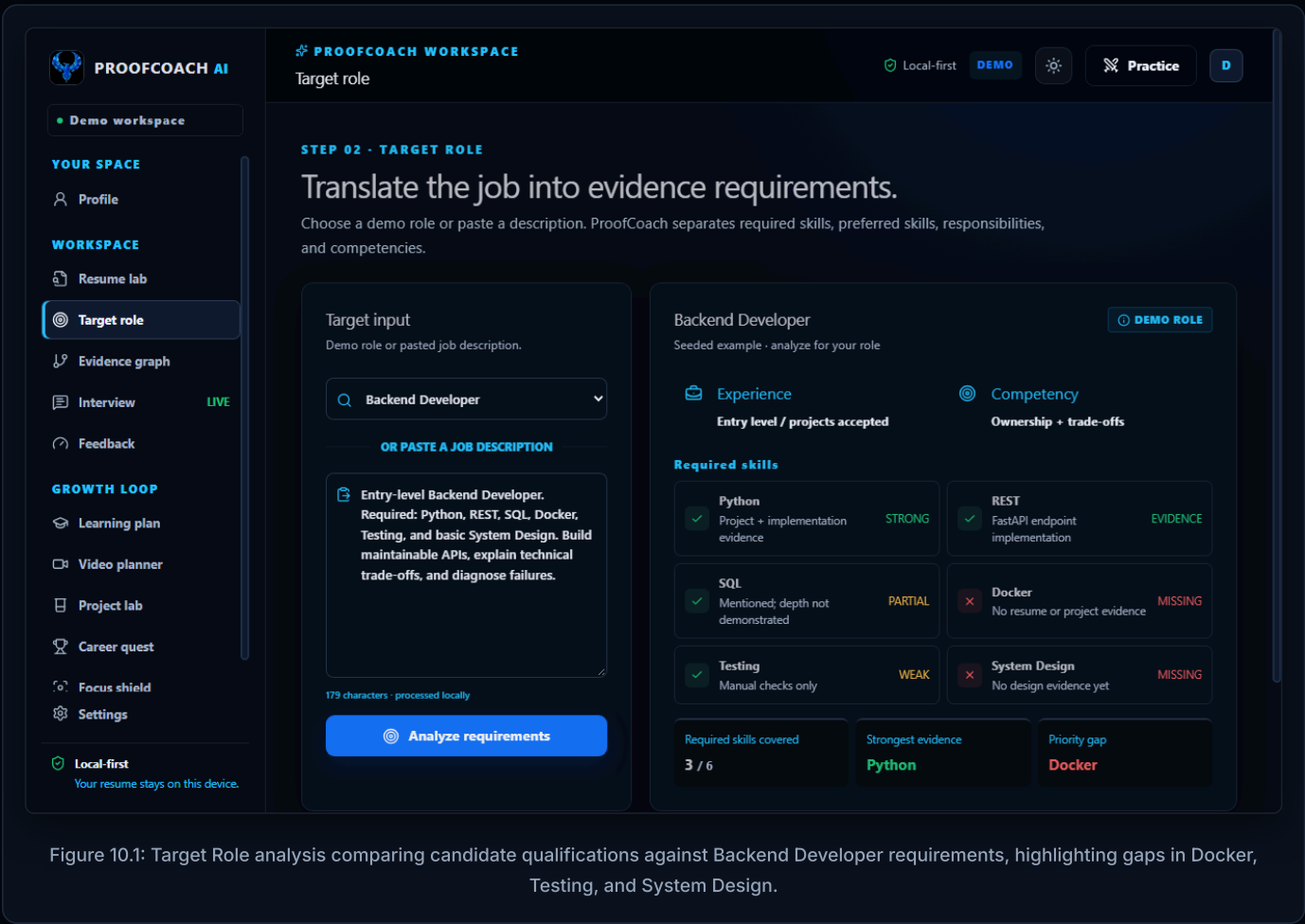


Figure 10.1: Target Role analysis comparing candidate qualifications against Backend Developer requirements, highlighting gaps in Docker, Testing, and System Design.

10.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The candidate selects a predefined demo role (e.g., Backend Developer, Frontend Developer, ML Engineer) or pastes a raw job description. The system instantly highlights which required skills the candidate already has evidence for (e.g., Python, REST, SQL) and marks missing skills as red gap indicators (e.g., Docker, Testing, System Design).

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

When the user submits a job description, `POST /api/role/analyze` searches the text against `DEMO_REQUIREMENTS`. If custom text is provided, regex word boundaries identify mentioned skills. The candidate's skills are compared case-insensitively

against the requirements. The system computes coverage percentage, identifies the strongest evidence area, and extracts the top 3 critical gaps to feed into the learning pipeline.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/RolePage.tsx` , `backend/app/services/evidence.py`
- **Functions:** `role_analyze()` , `analyze_job()` , `skill_coverage()`
- **Data Models:** `typescript interface RoleAnalysis { role: string; required_skills: string[]; preferred_skills: string[]; responsibilities: string[]; competencies: string[]; experience_requirement: string; }`
- **Coverage Status Logic:**
 - `strong` : Skill is present and part of primary evidence (e.g., Python, REST).
 - `evidence` / `partial` : Skill is detected in candidate profile with limited project depth.
 - `missing` : Skill is required by role but absent from candidate profile.

11. Career Evidence Graph System

11.1 Overview & Purpose

The Career Evidence Graph (`/evidence`) visualizes the candidate's professional claims as an interactive Directed Acyclic Graph (DAG). It implements the core principle that a resume bullet point is not verified evidence until it connects a candidate to a skill, a project, a measurable claim, an interview defense, and an explicit job requirement.

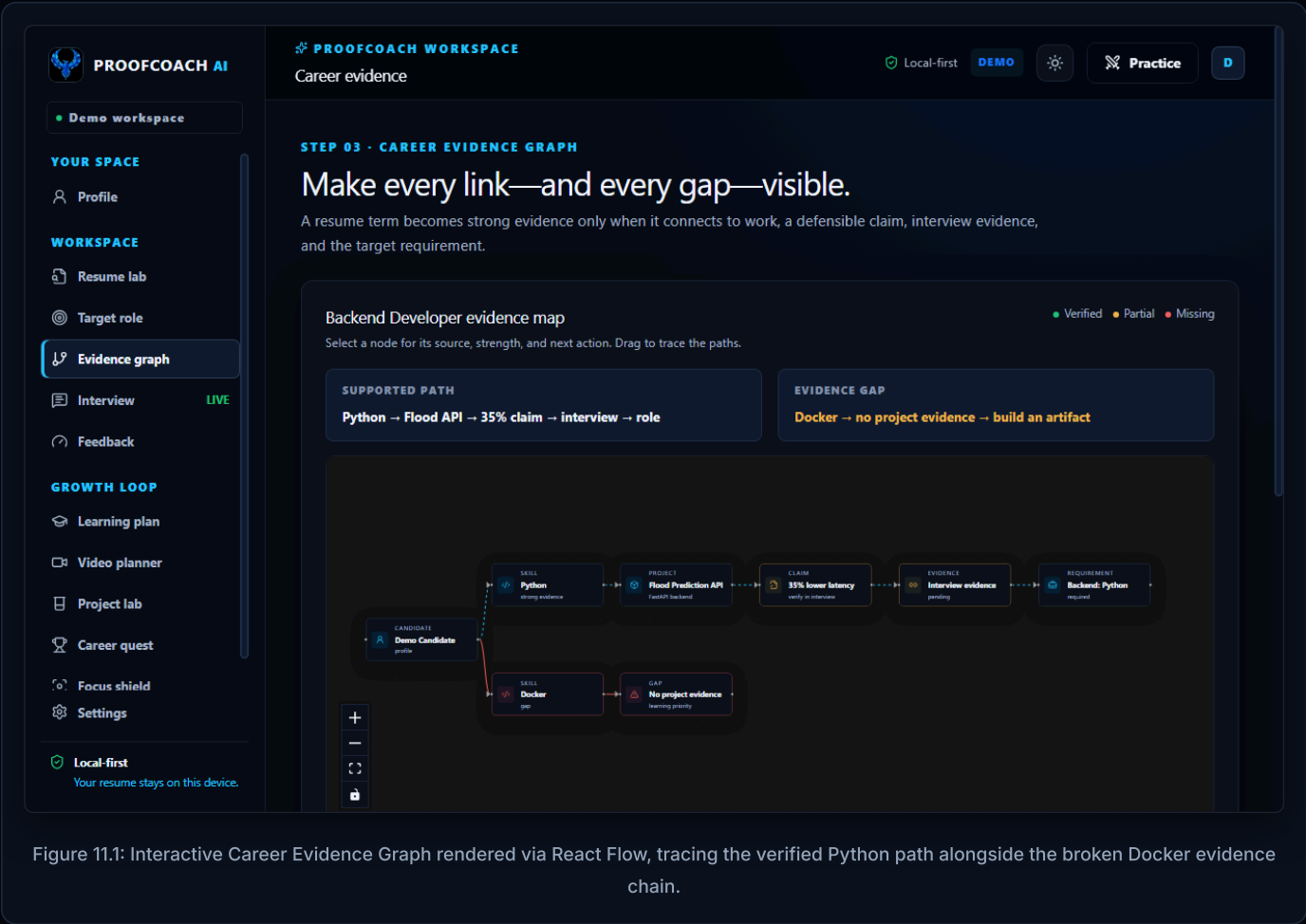


Figure 11.1: Interactive Career Evidence Graph rendered via React Flow, tracing the verified Python path alongside the broken Docker evidence chain.

11.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The user explores an interactive visual map with movable nodes. A green path traces supported evidence: Candidate → Python → Flood Prediction API → 35% Latency Claim → Interview Defense → Target Role. A red broken path highlights an evidence gap: Candidate → Docker → No Project Evidence → Gap. Clicking any node opens an Inspector sidebar detailing the node's source, evidence strength, and the exact next action needed.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

The graph is generated using `@xyflow/react` (React Flow). Node types are mapped to a custom `EvidenceNode` component. Node positions (x , y) are calculated to enforce a left-to-right progression across six horizontal stages. Edges connecting verified nodes are styled with crystal-blue animated strokes (`#27c4ff` or `#1769ff`), while edges connecting missing or unverified nodes are styled with red warning borders and static dashed markers.

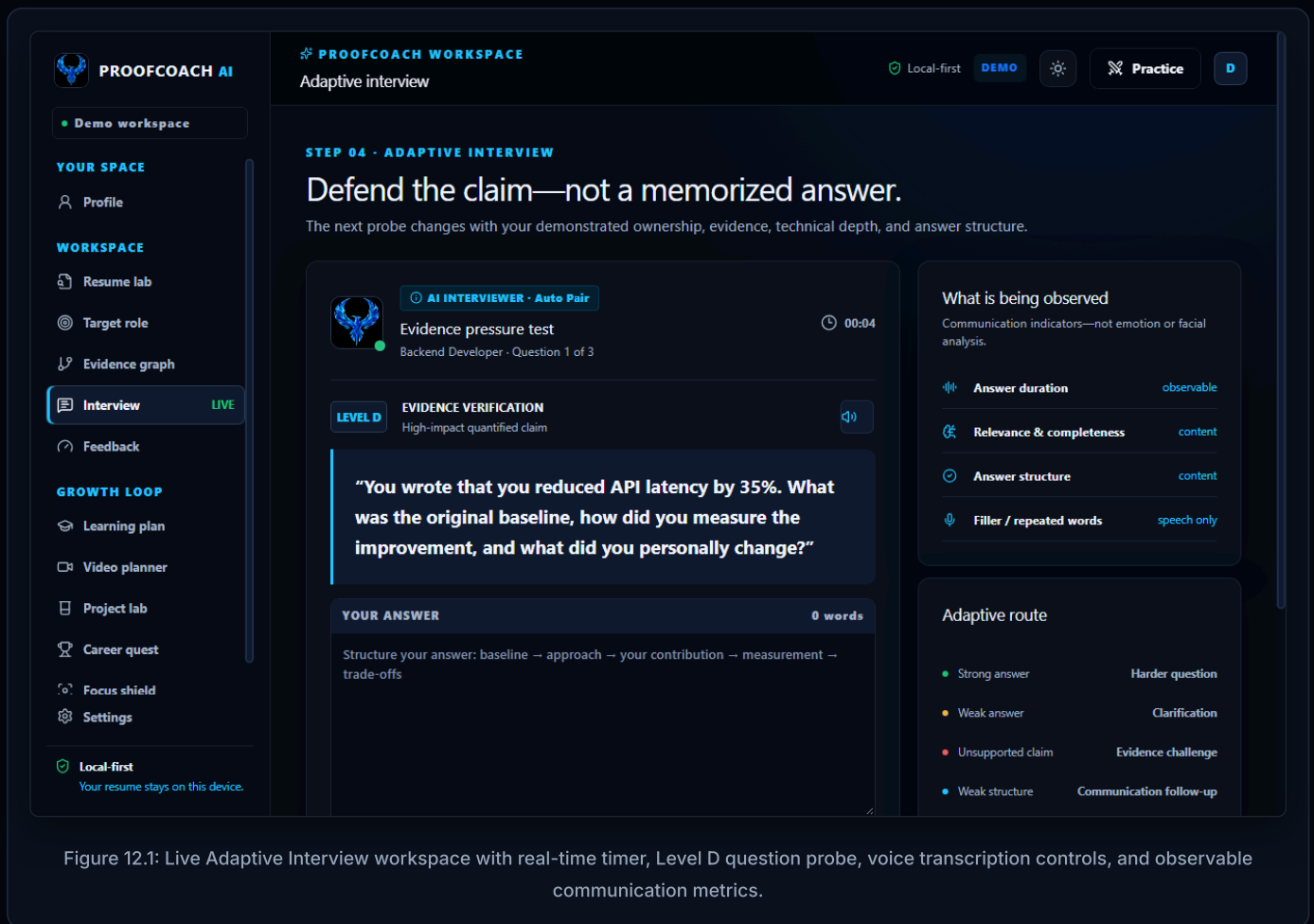
LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/EvidencePage.tsx` , `backend/app/services/evidence.py`
- **Graph Primitives:**
- **Nodes:** `c` (Candidate), `p` (Python), `d` (Docker), `project` (Flood API), `gap` (No project), `claim` (35% claim), `interview` (Interview session), `role` (Backend requirement).
- **Inspector State:** `selected: Node | null` triggers slide-in panel displaying `detail.kind` , `detail.status` , `detail.note` , and contextual next actions.
- **Color Modes:** Bound to `useTheme()` to ensure background grid dots and edge colors automatically adapt between dark navy and light editorial themes.

12. Live Adaptive Interview Engine

12.1 Overview & Purpose

The Live Adaptive Interview (`/interview`) provides a stateful, interactive environment where candidates are interrogated on their specific resume claims. It completely rejects canned, non-contextual interview scripts, instead deploying an adaptive probing progression from Level A through Level F.



12.2 The Six-Level Adaptive Probing Taxonomy

```
stateDiagram-v2
    [*] --> LevelA: Candidate Answers
    LevelA: Level A – Clarification
    (Explain basic terminology)
    LevelB: Level B – Ownership
    (What was your personal contribution?)
    LevelC: Level C – Technical Depth
    (Explain internal architecture)
    LevelD: Level D – Evidence Verification
    (What was baseline & measurement?)
    LevelE: Level E – Trade-offs
    (Downsides & when to avoid approach)
    LevelF: Level F – Pressure / Counterfactual
    (System failure & 10x traffic scale)

    LevelD --> LevelE: Strong Answer (Clear baseline + ownership)
    LevelD --> LevelA: Weak Answer (Vague explanation)
    LevelE --> LevelF: Deep Trade-off Mastery
    LevelE --> LevelB: Vague Contribution
```

12.3 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The user faces an AI interviewer avatar. The current question challenges their 35% latency claim: "What was the original baseline, how did you measure the improvement, and what did you personally change?" The user can click a speaker icon to hear the question spoken aloud, and can respond either by typing into a structured text box or speaking via browser speech recognition. A word counter and session timer track progress.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

1. On component mount, the frontend calls `POST /api/interview/start` to register an active session in SQLite and fetch Question 0 (Level D).
2. The user speaks or types. If voice is activated, the Web Speech API (`webkitSpeechRecognition`) streams interim transcripts into local state.
3. Upon clicking "Submit evidence", the text is transmitted to `POST /api/interview/{session_id}/answer` .
4. The backend evaluation service parses the answer for baseline markers (`\d+\s*(?:ms|seconds?|%)`), measurement terms (`benchmark` , `postman` , `p95`), personal pronouns (`I changed` , `I implemented`), trade-off terms (`cache invalidation` , `memory cost`), and filler words (`um` , `uh` , `like`).
5. A detailed evaluation payload is returned, updating the session transcript and determining the next question level.

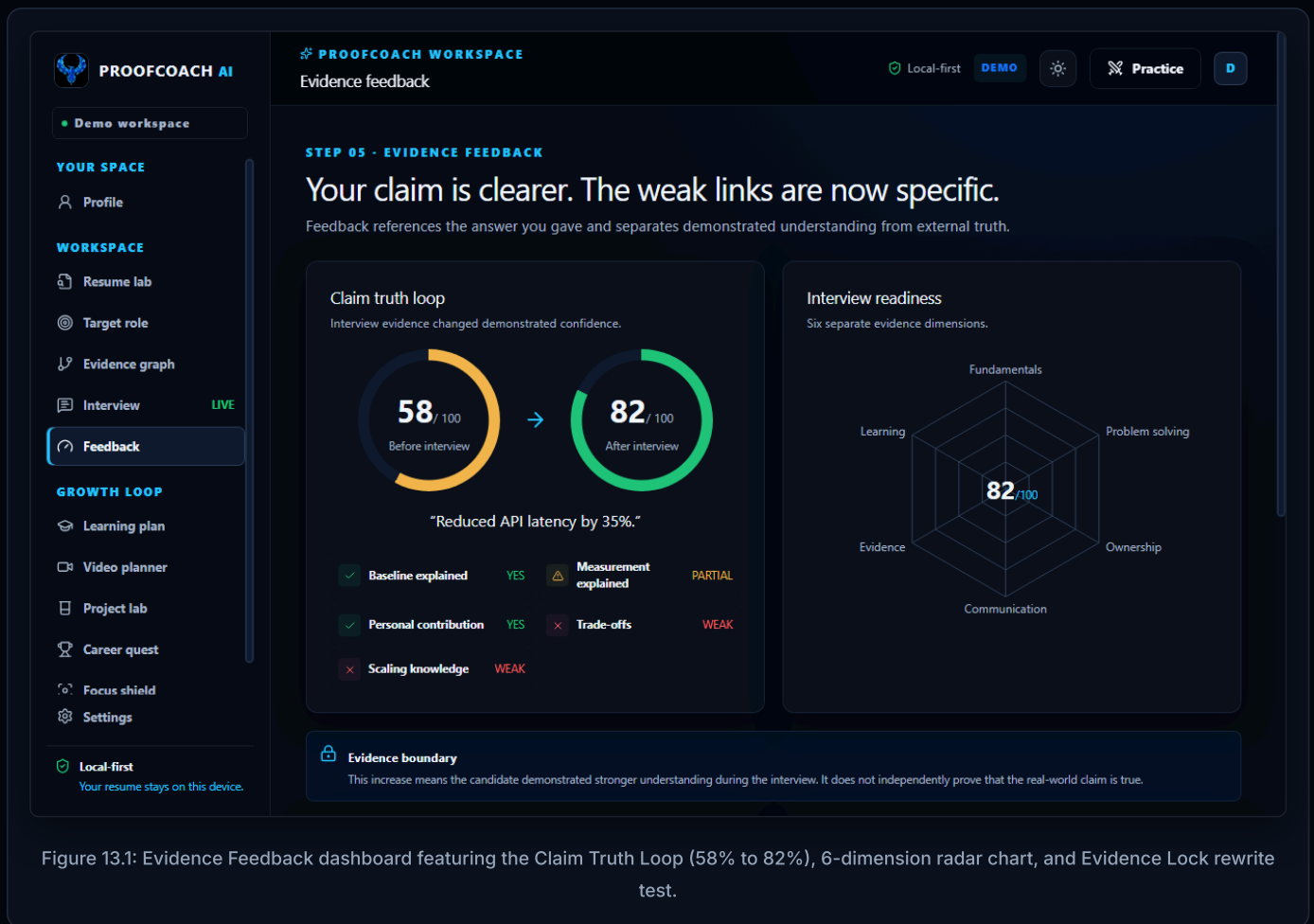
LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/InterviewPage.tsx` , `backend/app/services/interview.py` , `backend/app/main.py`
- **State Properties:** `state.answer: string` , `state.interviewSessionId: number | null` , `currentQ: { level, label, reason, text }`
- **Browser APIs Used:** `window.speechSynthesis` (for text-to-speech), `window.SpeechRecognition` / `window.webkitSpeechRecognition` (for voice input).
- **Communication Indicators Observed:**
 - `word_count` : Total tokens extracted via `\b[\w'-]+` .
 - `filler_words` : Count of `um` , `uh` , `like` , `basically` .
 - `answer_structure` : Classified as `clear` if ≥ 45 words with structured markers; otherwise `developing` .
 - `relevance` : Evaluated based on domain keywords (`latency` , `api` , `cache` , `measure`).

13. Evidence Feedback Engine & Evidence Lock

13.1 Overview & Purpose

The Feedback Engine (`/feedback`) delivers structured, multi-dimensional feedback following an interview. It features the **Resume Truth Loop**, which quantifies how the interview changed demonstrated claim confidence, and **Evidence Lock**, an interactive sandbox that protects candidates from generating fraudulent resume bullets.



13.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The screen displays a radar chart covering 6 dimensions: Fundamentals, Problem Solving, Ownership, Communication, Evidence, and Learning. The Resume Truth Loop shows that demonstrated confidence in the 35% latency claim increased from 58% to 82%. In the Evidence Lock box, the user can test two rewrites: a safe version ("Optimized API response performance") which is approved, and an unsupported version ("Reduced API latency by 40%") which is immediately blocked.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

The 82% claim confidence score does not claim independent real-world truth; it reflects the candidate's demonstrated ability to substantiate their baseline (420 ms) and measurement method (Postman runs). Evidence Lock enforces rules via `assess_rewrite()` in `backend/app/services/evidence_lock.py`: - Checks proposed text against confirmed numerical facts and technology keywords (`TECH_TERMS`). - If an unsupported number (e.g., `40%`) or technology (e.g., `Kubernetes`) appears in the rewrite without existing in the original bullet or confirmed facts, `status` returns `blocked` with a red UI token. - If only stylistic words change, `status` returns `safe` .

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

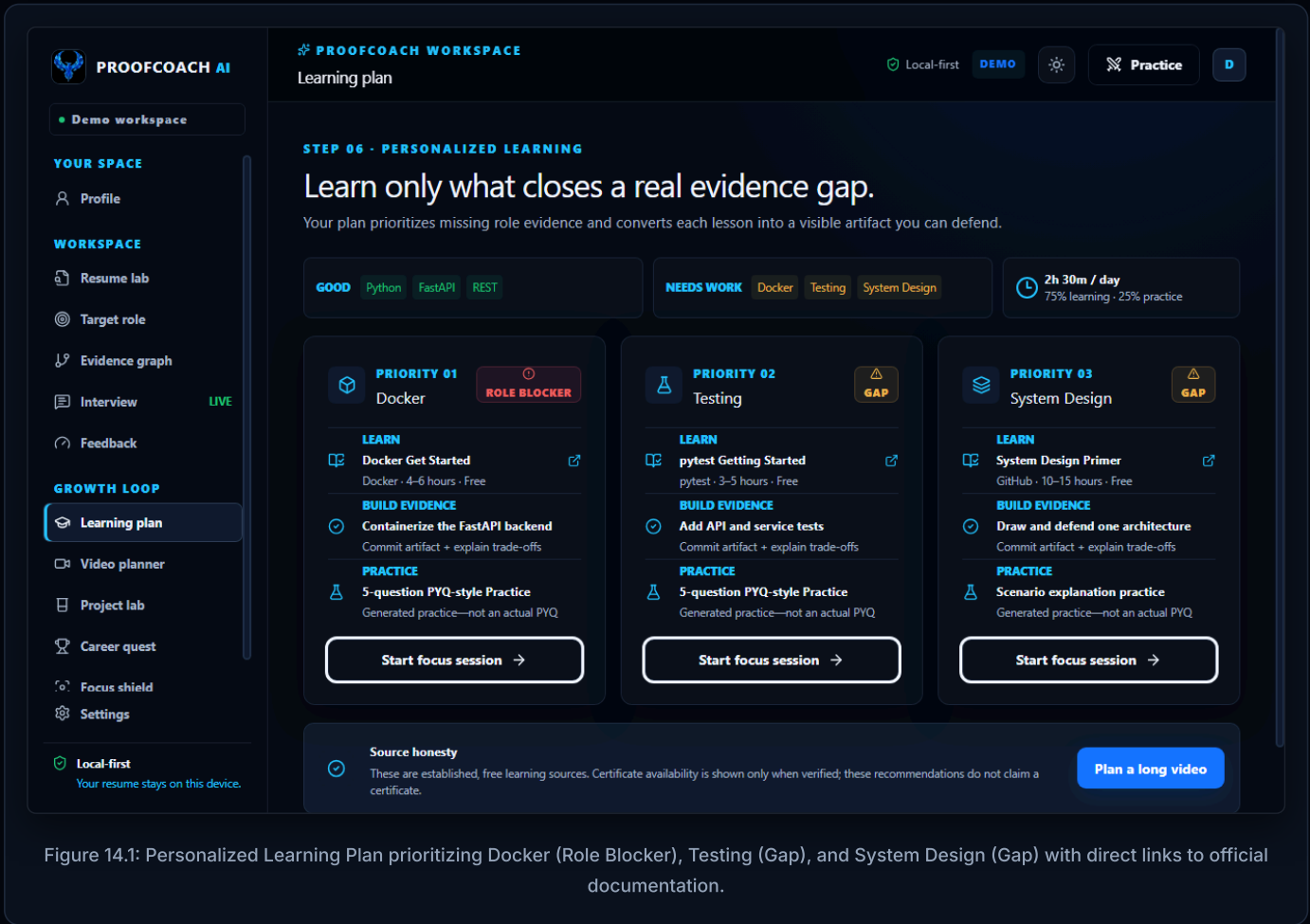
- **Source Files:** `frontend/src/pages/FeedbackPage.tsx` , `backend/app/services/evidence_lock.py`
- **Visual Primitives:** `<RadarChart>` wrapped inside `<ResponsiveContainer width="100%" height={285}>` .
- **Dimensions Evaluated:**

```
python dimensions = { "baseline": "yes" if baseline else "partial", "measurement": "yes" if measurement and has_digits else "partial" if measurement else "weak", "personal_contribution": "yes" if ownership else "partial", "trade_offs": "yes" if tradeoffs else "weak", "scaling_knowledge": "yes" if scaling else "weak", }
```
- **Confidence Formula:**
$$88 + 4 \times N_{\text{partial}} + \min(88, 58 + 8 \times N_{\text{weak}})$$

14. Personalized Learning Plan

14.1 Overview & Purpose

The Learning Plan (`/Learn`) closes the loop between interview failures and technical remediation. Instead of recommending 40-hour video courses, it generates concise, high-priority learning tracks targeting the candidate's exact role gaps.



14.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The user sees their target daily study time (2h 30m / day) divided into a 75% learning / 25% practice split. Three prioritized cards address their gaps: Priority 1 (Docker - Role Blocker), Priority 2 (Testing - Gap), and Priority 3 (System Design - Gap). Each card provides a direct link to free, authoritative documentation (e.g., Docker Get Started, pytest docs) and assigns a mandatory artifact to build.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

The component dispatches `POST /api/learning/plan` with the candidate's identified gaps and daily time budget. The backend maps each gap to verified primary resources in `RESOURCE_CATALOG`. Crucially, every task enforces an active triad: 1. **Learn:** Study official documentation. 2. **Build Evidence:** Create a concrete code artifact (e.g., "Containerize the FastAPI backend"). 3. **Practice:** Complete a 5-question PYQ-style practice set.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

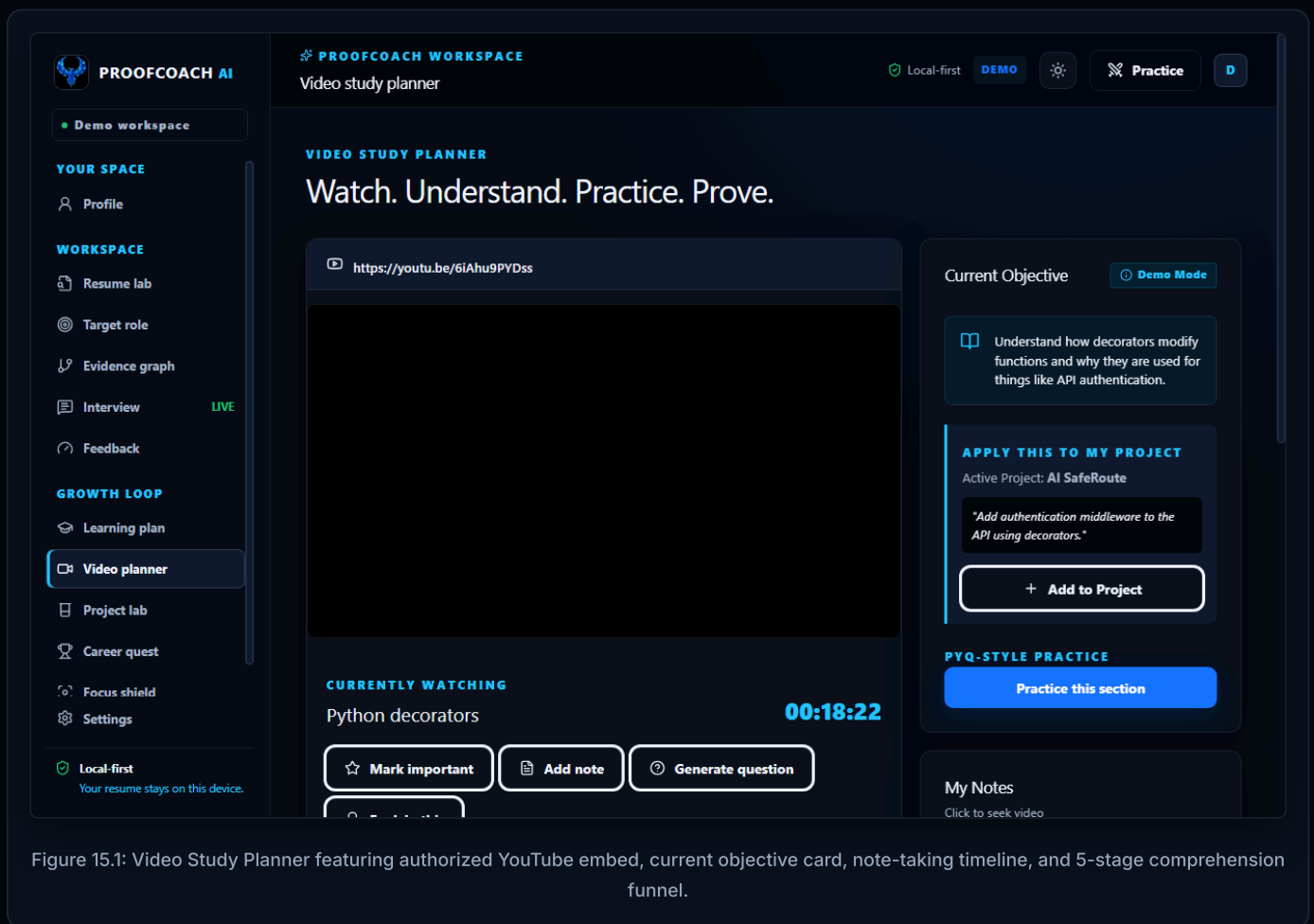
- **Source Files:** `frontend/src/pages/LearnPage.tsx`, `backend/app/services/Learning.py`

- **State Binding:** Dynamic time allocation calculated as: `typescript const hours = Math.floor(state.candidate.dailyMinutes / 60); const mins = state.candidate.dailyMinutes % 60;`
- **Source Honesty Policy:** Every external resource explicitly displays its provider, estimated time, and cost. If a certificate is not verified, the UI states: "These recommendations do not claim a certificate."

15. Video Study Planner

15.1 Overview & Purpose

The Video Study Planner (`/video-plan`) tackles the pervasive problem of "tutorial hell" by converting passive video consumption into an active, structured study schedule. It enforces an authorized YouTube embed player, daily video chunking, and timestamped note-taking.



15.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The candidate pastes a YouTube tutorial link. The planner embeds the video using privacy-enhanced mode (`youtube-nocookie.com`). As they watch, they can click "Add note" or "Mark important" to save timestamped notes. A 5-stage comprehension funnel tracks progress across: Watched (45%) → Understood (30%) → Practiced (15%) → Applied (10%) → Proven (5%). A daily planner breaks a multi-hour tutorial into manageable day-by-day segments.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

The component validates the URL using `videoId()` in `services/video.ts`, strictly requiring an 11-character alphanumeric video ID from `youtu.be` or `youtube.com`. Daily sessions are computed using the formula: $\text{ext}\{\text{Learning Budget}\} = \text{ext}\{\text{round}\}(\text{ext}\{\text{Daily Minutes}\} \times 0.75) \times \text{ext}\{\text{Days Required}\} = \lceil \text{ext}\{\text{Duration}\} / \text{ext}\{\text{Learning Budget}\} \rceil$ Notes taken during playback are committed to `state.video.notes` in `ProofContext` and survive page refreshes.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/VideoPlanPage.tsx`, `frontend/src/services/video.ts`, `backend/app/services/learning.py`
- **State Data Model:**

```
typescript interface VideoNote { id: string; timestamp: number; topic: string; text: string; } interface VideoState { watched: number; understood: number; practiced: number; applied: number; proven: number; notes: VideoNote[]; }
```
- **Copyright & Privacy Boundary:** The application never downloads, scrapes, or stores copyrighted video streams. It uses the official YouTube iframe embed with `referrerPolicy="strict-origin-when-cross-origin"`.

16. Project Lab: Artifact Engineering

16.1 Overview & Purpose

The Project Lab (`/project-lab`) is ProofCoach AI's execution engine. It ensures that students and freshers do not merely accumulate theoretical knowledge, but build, test, and document real software artifacts that can withstand technical scrutiny.

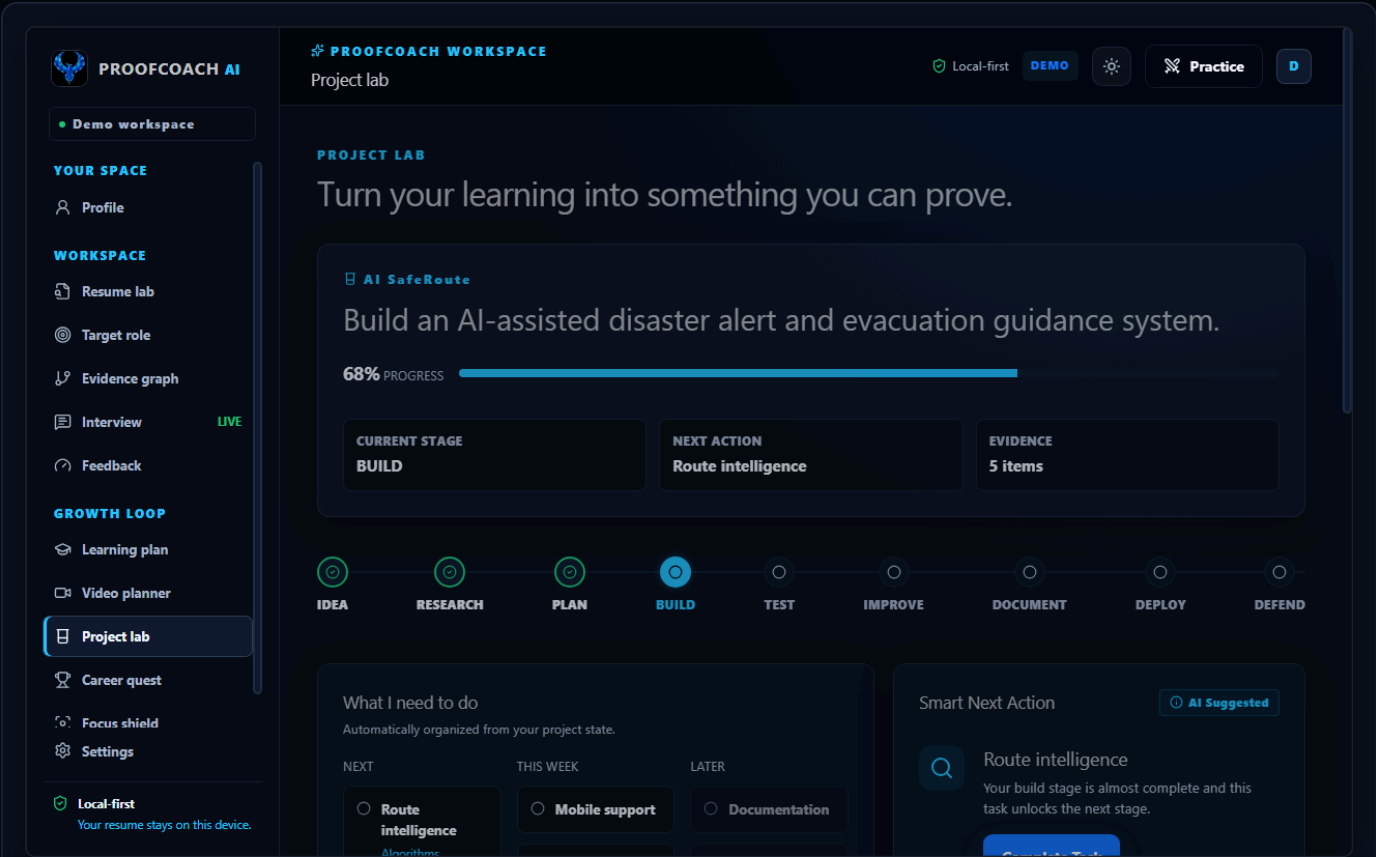
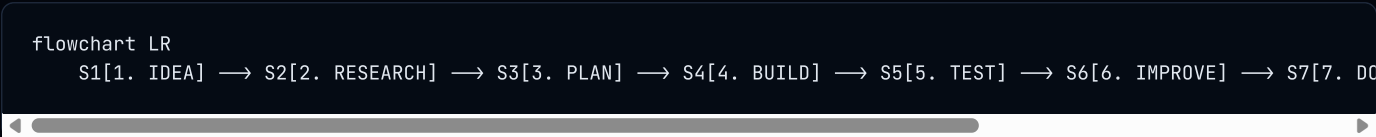


Figure 16.1: Project Lab workspace displaying the 9-stage engineering journey, dynamic task board (NEXT / THIS WEEK / LATER), and interview defense preview.

16.2 The Nine-Stage Engineering Journey

Every project progresses through an industry-standard development lifecycle:



16.3 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The user manages their active project (e.g., "AI SafeRoute - Flood Prediction API"). An engineering progress bar shows completion (24%). A Kanban task board divides actions into NEXT, THIS WEEK, LATER, and COMPLETED. Clicking any task marks it complete, immediately advancing project progress. A "Defend This Project" panel previews three core questions that an interviewer will ask about the project.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

When a candidate checks off a task, `completeTask(taskId)` is dispatched to `ProofContext`. The task's stage transitions to `COMPLETED`, its `completed` boolean flag is set to `true`, and overall project progress advances incrementally: $\text{ext}\{\text{Progress}\} \leftarrow \min(100, \text{ext}\{\text{Progress}\} + 4)$. Completed tasks with attached evidence strings automatically populate the "Project Evidence" panel, providing one-click export into the Resume Lab.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/ProjectLabPage.tsx`, `frontend/src/types.ts`
- **State Data Model:**

```
typescript interface ProjectTask { id: string; title: string; stage: 'NEXT' | 'THIS WEEK' | 'LATER' | 'COMPLETED'; completed: boolean; priority?: string; evidence?: string; skill?: string; } interface ProjectState { id: string; name: string; goal: string; stage: string; progress: number; tasks: ProjectTask[]; }
```
- **New Project Wizard:** If `state.project` is null, `ProjectLabPage` renders an embedded two-step setup wizard allowing the candidate to initialize a custom project with custom objectives.

17. Career Quest Progression Engine

17.1 Overview & Purpose

Career Quest (`/quest`) gamifies the career preparation journey through transparent, merit-based Evidence XP. It explicitly avoids dark-pattern gamification (e.g., daily login streaks or superficial clicks), rewarding only verified, artifact-backed achievements.

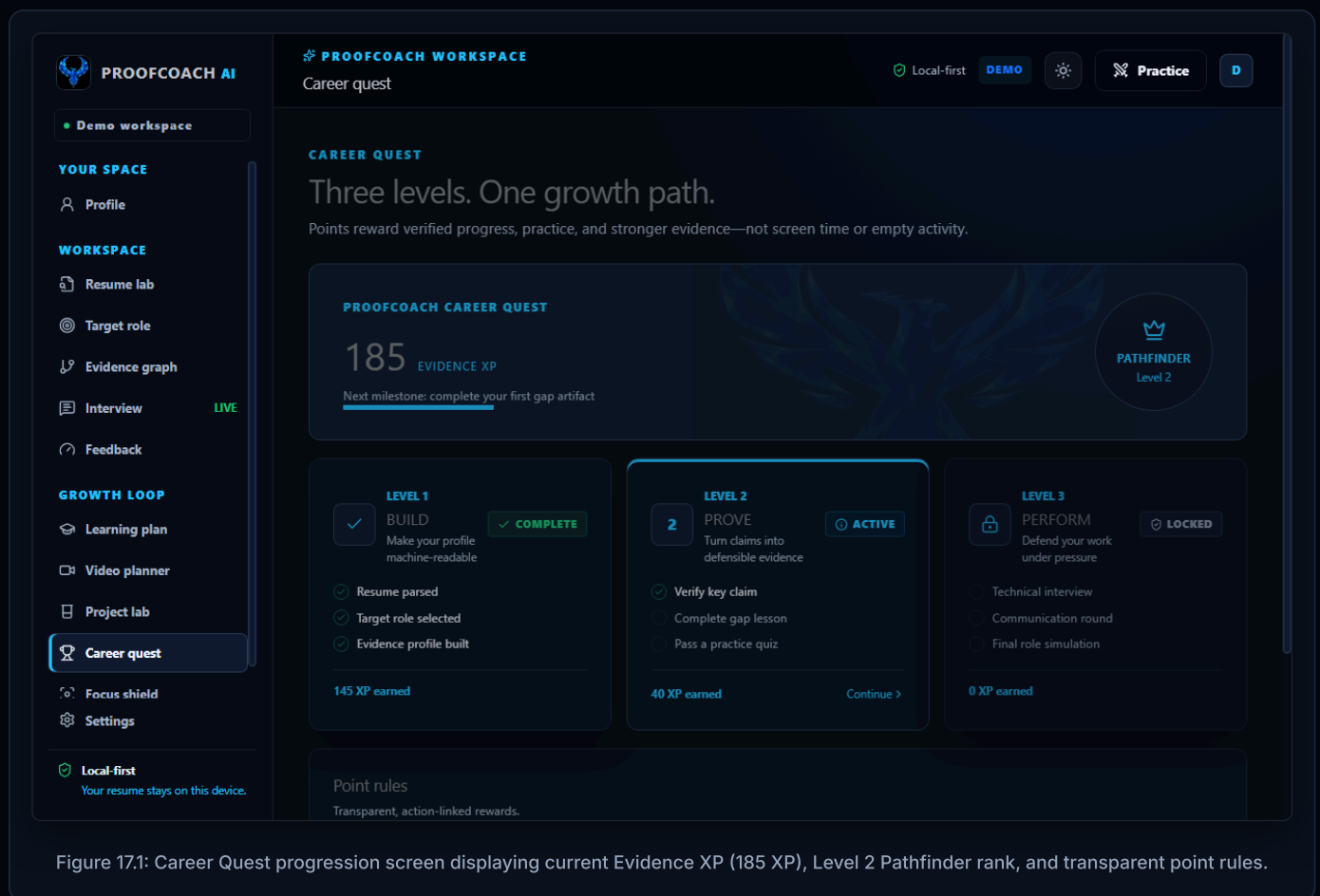


Figure 17.1: Career Quest progression screen displaying current Evidence XP (185 XP), Level 2 Pathfinder rank, and transparent point rules.

17.2 The Three Quest Tiers & XP Rules

QUEST LEVEL	NAME	THEME	KEY INCLUDED WORK	LEVEL REWARD
Level 1	Build	Foundation	Upload resume, fix parser warnings, select role, map evidence graph.	100 XP
Level 2	Prove	Competency	Verify claims, complete learning modules, build gap artifacts, Evidence Lock rewrite.	150 XP
Level 3	Perform	Mastery	Live interview defense, deep trade-off interrogation, final simulated interview.	250 XP

TRANSPARENT ACTION-LINKED REWARDS:

- Resume fix: +20 XP
- Evidence added: +25 XP
- Lesson completed: +10 XP
- Quiz passed: +15 XP
- Adaptive Interview completed: +30 XP (seeded demo awards +50 XP for comprehensive defense)
- Claim improved: +20 XP
- Complete tier milestone: +100 XP

17.3 Three-Layer Analysis

- **Level 1 (User View):** The user tracks their Evidence XP (185 XP) and rank ("Pathfinder - Level 2"). Level 1 is marked Complete with green checkmarks, Level 2 is Active, and Level 3 is Locked until Level 2 tasks are fulfilled.
- **Level 2 (Internal Logic):** XP is stored in `state.metrics.questPoints`. Progression percentage is derived as `questPoints / 4`. Submitting an interview automatically increments `questPoints` by 50 in `submitInterview()`.
- **Level 3 (Developer View):** Defined in `frontend/src/pages/QuestPage.tsx` and seeded via `questLevels` in `frontend/src/data/demo.ts`.

18. Focus Shield & Wellbeing System

18.1 Overview & Purpose

Focus Shield (`/focus`) provides an intensive, distraction-minimized study environment with an integrated wellbeing governor. It ensures that candidates maintain healthy learning habits by enforcing a 2.5-hour maximum continuous study session with automatic progress autosaving.

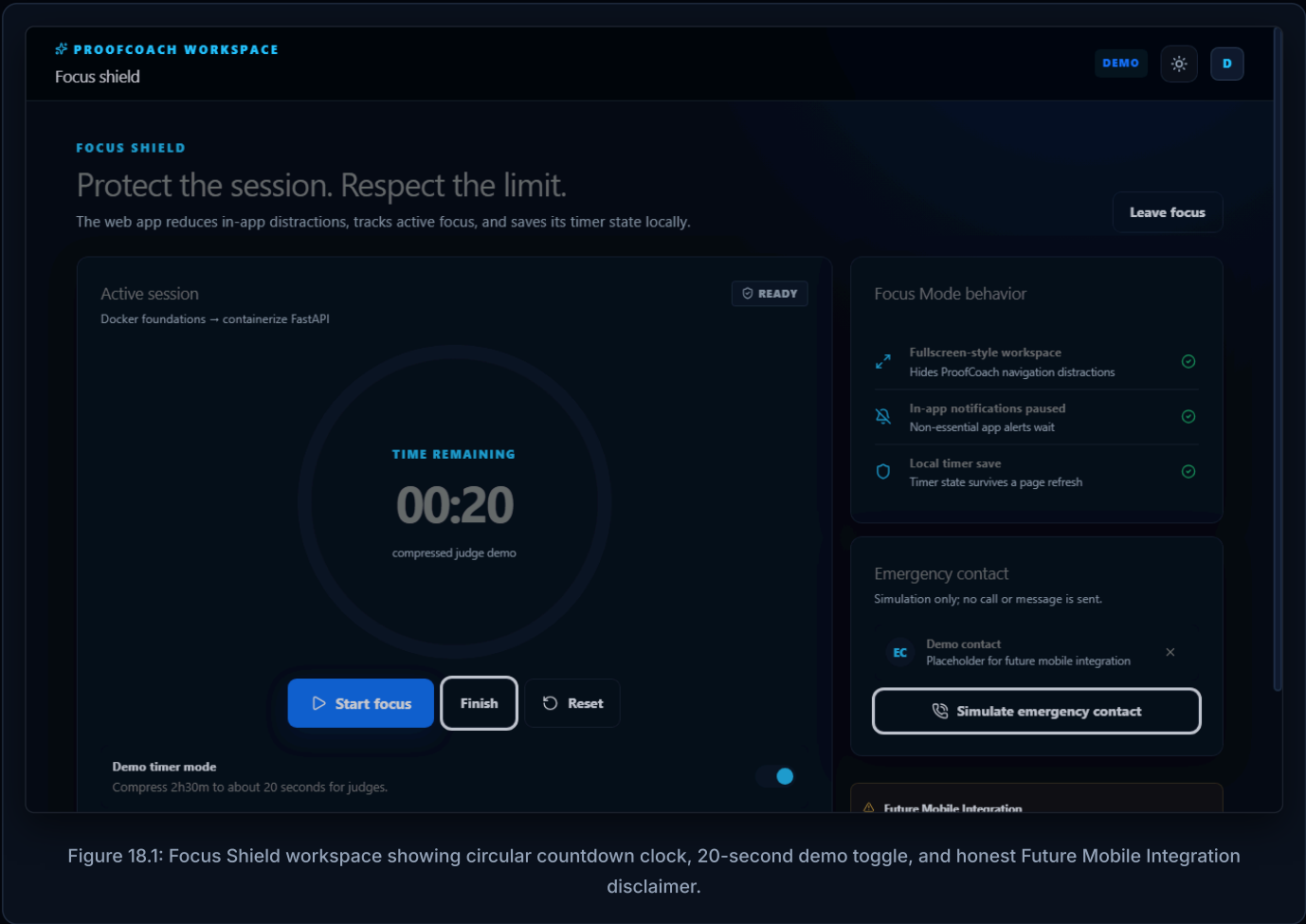


Figure 18.1: Focus Shield workspace showing circular countdown clock, 20-second demo toggle, and honest Future Mobile Integration disclaimer.

18.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The screen strips away distracting navigation elements, presenting a large circular countdown timer set to a 2h 30m wellbeing limit. Controls allow the user to Start, Pause, Finish, or Reset the session. For quick demonstrations, a "Demo timer mode" toggle compresses the 2.5-hour limit to 20 seconds. An Emergency Contact simulator is provided, accompanied by an honest notice explaining that web browsers cannot block phone calls.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

The timer runs client-side via a 1000 ms `setInterval` loop. The CSS conical gradient (`--progress`) dynamically visualizes elapsed percentage: $\frac{\text{Progress}}{\text{Limit}} \times 100$ When `elapsed ≥ limit`, the session ends automatically, `running` is set to `false`, `ended` is set to `true`, and state is autosaved to `localStorage` under `proofcoach-focus-v1`.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/FocusPage.tsx`, `backend/app/services/focus.py`
- **Storage Key:** `proofcoach-focus-v1` storing `{ demo: boolean, elapsed: number, ended: boolean }`.
- **Localhost Optimization:** To eliminate unnecessary network round-trips every second, the timer loop executes entirely in the browser. The backend endpoint `POST /api/focus/status` remains available for server-side state synchronization.
- **Honest System Boundary:** The UI features an explicit warning: "A browser website cannot guarantee blocking WhatsApp, phone calls, or OS notifications. This prototype does not fake that capability."

19. Settings, Runtime Controls & Privacy Purge

19.1 Overview & Purpose

The Settings module (`/settings`) implements ProofCoach AI's core privacy and data ownership guarantees. It provides instant theme switching, runtime mode inspection, complete data export, and an irreversible one-click data destruction mechanism.

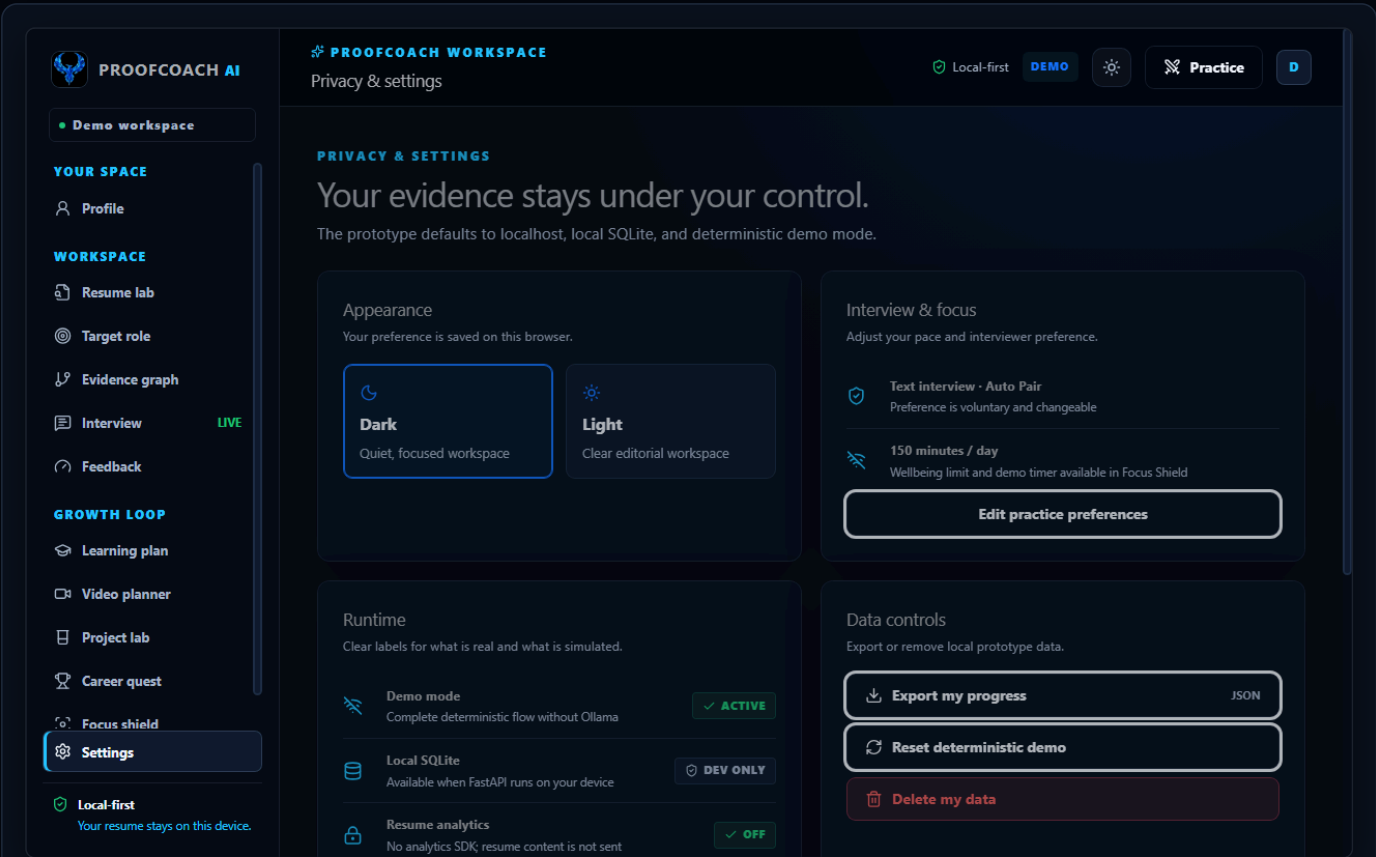


Figure 19.1: Settings and Privacy panel showing appearance options, runtime diagnostic indicators, JSON export, and Delete My Data purge modal.

19.2 Three-Layer Analysis

LEVEL 1 — SIMPLE EXPLANATION (USER VIEW)

The user can toggle between Dark Navy and White Editorial themes. Runtime cards confirm that the system is operating in Local-First Demo Mode with zero analytics tracking. Three action buttons allow the candidate to export their entire progress as a JSON file, reset the deterministic demo to its original state, or permanently delete all local data.

LEVEL 2 — INTERNAL EXPLANATION (SYSTEM LOGIC)

1. **JSON Export:** Serializes the complete reactive `DemoState` object into a formatted JSON Blob and triggers an automated browser download for `proofcoach-progress.json`.
2. **Deterministic Reset:** Calls `resetDemo()` in `ProofContext`, replacing the local state with `initialDemoState`.
3. **Data Purge Cascade:** Clicking "Delete my data" opens an accessible confirmation modal. Upon confirmation, the application dispatches `DELETE /api/data` to the FastAPI backend (which truncates all rows in `CandidateProfile`, `ResumeRecord`, and `InterviewSession`), purges `localStorage` keys `proofcoach-demo-v1` and `proofcoach-focus-v1`, and navigates the user back to the landing page.

LEVEL 3 — DEVELOPER VIEW (TECHNICAL SPECIFICATION)

- **Source Files:** `frontend/src/pages/SettingsPage.tsx`, `frontend/src/services/api.ts`, `backend/app/main.py`
- **Data Purge Implementation:**

```
python @app.delete("/api/data") def delete_my_data(session: Session = Depends(get_session)) → dict: for model in (InterviewSession, ResumeRecord, CandidateProfile): session.exec(delete(model)) session.commit() return {"deleted": ["profile", "resume", "interview transcript"], "local_database_retained_empty": True}
```

20. Frontend Architecture & UI Design System

20.1 Directory Layout & Component Structure

The frontend is constructed using React 19, TypeScript 5.9, and Vite 7. The project maintains an intentional separation between reusable UI primitives, global shell layouts, route-level page views, and state providers.

```

frontend/
├─ public/
│  └─ assets/
│     ├── phoenix-watermark.jpg    # Official Blue Crystal Phoenix asset
│     └─ favicon.svg
├─ src/
│  ├── components/                # Shared UI primitives & Layout Shell
│  │  ├── Brand.tsx              # Phoenix logo & typography lockup
│  │  ├── Shell.tsx              # Global responsive sidebar, topbar & navigation
│  │  ├── ThemeToggle.tsx        # Accessible light/dark theme switch
│  │  └─ UI.tsx                  # Atomic design tokens (Panel, ScoreRing, StatusPill)
│  ├── context/                  # React Context providers
│  │  ├── ProofContext.tsx       # Central application state & persistence
│  │  └─ ThemeContext.tsx        # Dual-theme engine & preference sync
│  ├── data/
│  │  └─ demo.ts                 # Deterministic fixtures & initial demo state
│  ├── pages/                    # Lazy-loaded route components (13 total)
│  ├── services/                 # Client-side utility functions & HTTP client
│  │  ├── api.ts                 # Fetch wrapper targeting localhost:8000
│  │  ├── theme.ts               # Theme resolution & local storage engine
│  │  └─ video.ts                # YouTube URL parser & ID extractor
│  ├── App.tsx                   # React Router configuration & route definitions
│  ├── main.tsx                  # DOM entry point & provider bootstrapping
│  ├── styles.css                 # Semantic CSS variables & responsive rules
│  └─ types.ts                   # Strict TypeScript domain interfaces
├─ package.json
└─ vite.config.ts

```

20.2 Design System & Dual-Theme Tokens

The interface utilizes a technical, crystal-inspired aesthetic. It supports two full-palette themes powered by CSS custom properties defined in `styles.css`.

SEMANTIC VARIABLE	DARK NAVY THEME (DEFAULT)	WHITE EDITORIAL THEME	SEMANTIC ROLE
<code>--bg</code>	<code>#03060D</code> (Deep Void Navy)	<code>#F7F9FC</code> (Cool Editorial White)	Application canvas background
<code>--surface</code>	<code>#0A111D</code> (Glass Card Surface)	<code>#FFFFFF</code> (Pure Crisp White)	Card and panel containers
<code>--surface-subtle</code>	<code>#0F172A</code>	<code>#F1F5F9</code>	Table rows, code blocks, dividers
<code>--text</code>	<code>#F4F8FF</code> (High-Contrast White)	<code>#111827</code> (Deep Editorial Slate)	Primary headlines and body copy
<code>--muted</code>	<code>#7E8FA6</code>	<code>#64748B</code>	Subtitles, timestamps, placeholders
<code>--primary</code>	<code>#1877FF</code> (Vibrant Electric Blue)	<code>#1769FF</code> (Refined Ink Blue)	Primary action buttons and focus rings
<code>--crystal</code>	<code>#27C4FF</code> (Luminous Cyan)	<code>#0070B0</code> (Deep Precision Cyan)	Brand accents, active meters, graph nodes
<code>--success</code>	<code>#22C77A</code> (Verified Emerald)	<code>#16875F</code> (Forest Green)	Passing scores, verified claims
<code>--warning</code>	<code>#F2B84B</code> (Amber Alert)	<code>#A66413</code> (Warm Ochre)	Partial evidence, layout warnings
<code>--danger</code>	<code>#F26161</code> (Critical Coral)	<code>#D64545</code> (Crimson Warning)	Blocked claims, role blockers

PREVENTING FLASH OF UNSTYLED THEME (FOUC):

In `index.html`, an inline bootstrap script executes synchronously before React mounts:

```
const saved = localStorage.getItem('proofcoach-theme');
const theme = saved === 'light' || saved === 'dark' ? saved : 'dark';
document.documentElement.setAttribute('data-theme', theme);
```

21. Backend Architecture & Service Layer

21.1 Module Structure & Separation of Concerns

The backend is built with FastAPI and Python 3.11+. It follows a pure-function domain service pattern where business logic is decoupled from HTTP transport.

```

backend/
├─ app/
│  ├─ config.py           # Pydantic Settings (CORS, demo mode, ports)
│  ├─ db.py              # SQLite database engine & session dependency
│  ├─ main.py            # FastAPI application, route handlers & lifespan
│  ├─ models.py          # SQLAlchemy persistence entities (table=True)
│  ├─ schemas.py         # Pydantic v2 request/response schemas
│  ├─ seed.py            # Deterministic resume text & state fixtures
│  ├─ providers/         # AI Provider abstraction layer
│  │  ├─ base.py         # Abstract base class for AI operations
│  │  └─ demo.py        # Deterministic implementation (zero external calls)
│  └─ services/         # Pure domain logic functions
│     ├─ evidence.py     # Job matching, skill coverage & graph builder
│     ├─ evidence_lock.py # Regex & dictionary claim classification
│     ├─ focus.py        # Session timer & wellbeing logic
│     ├─ interview.py    # Adaptive questioning & rubric evaluation
│     ├─ learning.py     # Gap-to-resource curriculum generator
│     └─ resume.py       # PyMuPDF text & layout extraction
├─ tests/
│  └─ test_core.py       # 9 automated unit tests verifying core logic
├─ requirements.txt
└─ pytest.ini

```

21.2 The Provider Abstraction Pattern

To support both zero-network hackathon demonstrations and optional self-hosted LLMs, the platform defines a strict provider interface in `backend/app/providers/base.py`:

```

class AIProvider(ABC):
    @abstractmethod
    def extract_resume_facts(self, text: str) → dict: ...
    @abstractmethod
    def analyze_job_requirements(self, description: str) → dict: ...
    @abstractmethod
    def generate_interview_question(self, context: dict) → dict: ...
    @abstractmethod
    def evaluate_interview_answer(self, answer: str, context: dict) → dict: ...
    @abstractmethod
    def assess_evidence_lock_rewrite(self, original: str, proposed: str) → dict: ...

```

The active `DeterministicAIProvider` implements this interface using deterministic heuristics, ensuring the platform runs reliably offline without unexpected token costs or network latency.

22. State Management: ProofContext & Reactive Loops

22.1 Global State Architecture

Client-side state is centralized in `ProofContext.tsx` via the `useProof()` hook. It manages a unified `DemoState` interface that synchronizes user actions across all 13 routes.

```

flowchart TD
    subgraph Browser Storage
        LS["localStorage  
key: 'proofcoach-demo-v1'"]
    end

    subgraph ProofProvider Reactive Scope
        State["state: DemoState"]
        Mutators["State Mutator Actions  
(updateCandidate, completeTask, submitInterview)"]
        Commit["commit(nextState)"]

        Mutators --> Commit
        Commit --> State
        Commit --> LS
        LS -. Initial Load .-> State
    end

    subgraph Consuming Pages
        P1["ProfilePage"]
        P2["ResumePage"]
        P3["RolePage"]
        P4["InterviewPage"]
        P5["ProjectLabPage"]
        P6["QuestPage"]
    end

    State --> P1
    State --> P2
    State --> P3
    State --> P4
    State --> P5
    State --> P6

```

22.2 State Property Inventory & Mutation Rules

PROPERTY	DATA TYPE	DEFAULT / SEEDED VALUE	MUTATION ACTION	PERSISTENCE SCOPE
<code>candidate</code>	<code>Candidate</code>	Demo Candidate (BE AI/ML)	<code>updateCandidate(partial)</code>	<code>localStorage</code>
<code>completed</code>	<code>string[]</code>	<code>['profile', 'resume', ...]</code>	<code>complete(key)</code>	<code>localStorage</code>
<code>metrics</code>	<code>Record<MetricKey, number></code>	<code>{ parser: 82, coverage: 50, ... }</code>	<code>setMetric(key, val)</code>	<code>localStorage</code>
<code>interviewAnswered</code>	<code>boolean</code>	<code>false</code>	<code>submitInterview()</code>	<code>localStorage</code>
<code>currentQuestion</code>	<code>number</code>	<code>0</code> (Level D)	<code>submitInterview()</code>	<code>localStorage</code>
<code>answer</code>	<code>string</code>	<code>""</code>	<code>setAnswer(str)</code>	<code>localStorage</code>
<code>interviewSessionId</code>	<code>number \ null</code>	<code>null</code>	<code>startInterview(id)</code>	<code>localStorage</code>
<code>evaluation</code>	<code>any \ null</code>	<code>null</code>	<code>submitInterview(eval)</code>	<code>localStorage</code>
<code>evidenceLockDemo</code>	<code>'idle'\ 'safe'\ 'blocked'</code>	<code>'idle'</code>	<code>setEvidenceLockDemo(val)</code>	<code>localStorage</code>
<code>project</code>	<code>ProjectState \ null</code>	AI SafeRoute (Flood API)	<code>updateProject()</code> , <code>completeTask()</code>	<code>localStorage</code>
<code>video</code>	<code>VideoState</code>	<code>{ watched: 45, notes: [...] }</code>	<code>addVideoNote(note)</code>	<code>localStorage</code>

23. Data Flow Traces across System Boundaries

Flow A: The Resume → Evidence → Interview → Feedback Loop

flowchart LR

A["Uploaded Resume (PDF)"] -->|PyMuPDF Extract| B["35% Latency Claim"]

B -->|Build Graph| C["Evidence Graph DAG"]

C -->|Identify Gap| D["Docker Missing"]

C -->|Trigger Probe| E["Interview Question (Level D)"]

E -->|Evaluate Answer| F["Truth Loop: 58% → 82%"]

F -->|Enforce Gate| G["Evidence Lock Rewrite"]

G -->|Generate Plan| H["Docker & Testing Tasks"]

Flow B: Live Interview Execution & Evaluation Flow

```
sequenceDiagram
    participant UI as InterviewPage
    participant API as /api/interview/{id}/answer
    participant Svc as services.interview
    participant DB as SQLite DB
    participant Ctx as ProofContext

    UI->>API: POST { answer: "Baseline was 420 ms..." }
    API->>Svc: evaluate_answer(answer)
    Svc->>Svc: Regex pattern match (baseline, tools, fillers)
    Svc-->>API: Returns evaluation payload & scores
    API->>DB: Append transcript to InterviewSession
    API-->>UI: { evaluation, next_question }
    UI->>Ctx: submitInterview(evaluation)
    Ctx->>Ctx: Increment XP (+50), set metrics.claim = 82
    UI->>UI: Navigate to /feedback
```

24. Persistence, Storage & Data Lifecycle

ProofCoach AI enforces strict data governance across all storage mediums:

STORAGE MEDIUM	KEY / TABLE NAME	DATA STORED	LIFETIME	RESET / PURGE BEHAVIOR
Browser LocalStorage	proofcoach-demo-v1	Serialized <code>DemoState</code> JSON (candidate, metrics, project, video notes)	Persistent across sessions	Wiped on "Delete My Data" or reset to demo fixtures
Browser LocalStorage	proofcoach-focus-v1	Active timer state <code>{ demo, elapsed, ended }</code>	Persistent across refreshes	Wiped on "Delete My Data"
Browser LocalStorage	proofcoach-theme	Appearance string (<code>'dark'</code> or <code>'light'</code>)	Persistent across sessions	Preserved across demo resets
Local SQLite DB	candidateprofile	Candidate name, category, education, preferences	Persistent on host machine	Truncated via <code>DELETE /api/data</code>
Local SQLite DB	resumerecord	Extracted text preview, parser score, analysis JSON	Persistent on host machine	Truncated via <code>DELETE /api/data</code>
Local SQLite DB	interviewsession	Session level, question index, full transcript JSON	Persistent on host machine	Truncated via <code>DELETE /api/data</code>
RAM (In-Memory)	Fast API UploadFile	Raw PDF/DOCX binary stream (max 8 MB)	Lifetime of HTTP request only	Destroyed immediately after text extraction

25. APIs & Local Services Specification

The FastAPI backend exposes **13 high-performance REST endpoints** on `http://localhost:8000` :

```
flowchart LR
    subgraph REST_API_Surface [REST API Surface]
        H["H[/api/health GET]"]
        DR["DR[/api/demo/reset POST]"]
        DS["DS[/api/demo/state GET]"]
        RP["RP[/api/resume/parse POST]"]
        RA["RA[/api/role/analyze POST]"]
        EG["EG[/api/evidence/graph POST]"]
        CE["CE[/api/claims/extract POST]"]
        IS["IS[/api/interview/start POST]"]
        IA["IA[/api/interview/{id}/answer POST]"]
        EL["EL[/api/evidence-lock/rewrite POST]"]
        LP["LP[/api/learning/plan POST]"]
        VP["VP[/api/video/plan POST]"]
        DD["DD[/api/data DELETE]"]
    end
```

Complete OpenAPI-Style Endpoint Index

METHOD	ENDPOINT PATH	REQUEST BODY	RESPONSE SCHEMA	DESCRIPTION
GET	/api/health	None	<code>{ status, mode, local_first }</code>	Healthcheck and runtime provider verification.
POST	/api/demo/reset	None	<code>{ reset, state }</code>	Truncates SQLite tables and re-seeds deterministic demo data.
GET	/api/demo/state	None	<code>DemoState</code> JSON	Returns full deterministic view model for zero-setup demos.
POST	/api/resume/parse	<code>multipart/form-data (file)</code>	<code>ParserAnalysis</code>	Extracts text and evaluates the 8-point robustness rubric.
POST	/api/role/analyze	<code>{ role, job_description }</code>	<code>RoleAnalysis</code>	Analyzes role requirements, responsibilities, and competencies.
POST	/api/evidence/graph	<code>{ candidate_skills, required_skills }</code>	<code>{ nodes, edges, coverage }</code>	Constructs DAG node and edge topology for React Flow.
POST	/api/claims/extract	<code>{ text }</code>	<code>{ claims: [...] }</code>	Extracts quantified achievement statements from raw text.
POST	/api/interview/start	<code>{ claim, mode }</code>	<code>{ session_id, level, question }</code>	Initializes interview session record in SQLite.
POST	/api/interview/{id}/answer	<code>{ answer }</code>	<code>{ evaluation, next_question }</code>	Scores candidate answer and determines next adaptive probe.
POST	/api/evidence-lock/rewrite	<code>{ original, proposed, confirmed }</code>	<code>{ status, color, message, ... }</code>	Classifies rewrite into

METHOD	ENDPOINT PATH	REQUEST BODY	RESPONSE SCHEMA	DESCRIPTION
				Safe, Confirm, or Blocked.
POST	/api/learning/plan	{ gaps, daily_minutes }	{ daily_minutes, split, tasks }	Generates prioritized learning curriculum and artifact goals.
POST	/api/video/plan	{ title, duration, daily_minutes, notes }	{ days, sessions, concepts }	Chunks long video tutorials into daily study/practice sessions.
DELETE	/api/data	None	{ deleted, local_database_retained_empty }	Permanent purge of all candidate database records.

26. Algorithms & Internal Logic Implementation

26.1 Parser Robustness Scoring Formula

The robustness score S_{robust} is calculated as: $S_{\text{robust}} = \max(0, \min(100, \sum_{i=1}^8 P_i))$ Where P_i represents points awarded for each of the 8 checks: 1. $P_1 = \text{round}((20 \times \min(1.0, L_{\text{text}} / 675)))$ 2. $P_2 = 10 \text{ if (email and phone) else } 5$ 3. $P_3 = \min(15, N_{\text{sections}} \times 3)$ 4. $P_4 = 10 \text{ if dates found else } 4$ 5. $P_5 = \min(15, N_{\text{skills}} \times 3)$ 6. $P_6 = 10 \text{ if projects found else } 0$ 7. $P_7 = 4 \text{ if two-column else } 10$ 8. $P_8 = 4 \text{ if tables found else } 10$

26.2 Evidence Lock Verification Logic

```
def assess_rewrite(original: str, proposed: str, confirmed_facts: list[str] = None):
    confirmed_facts = confirmed_facts or []
    # 1. Extract numerical tokens
    original_numbers = set(NUMBER_PATTERN.findall(original)) | {
        item for fact in confirmed_facts for item in NUMBER_PATTERN.findall(fact)
    }
    proposed_numbers = set(NUMBER_PATTERN.findall(proposed))
    invented_numbers = sorted(proposed_numbers - original_numbers)

    # 2. Extract technical terms
    original_tech = {t for t in TECH_TERMS if t.lower() in original.lower()}
    proposed_tech = {t for t in TECH_TERMS if t.lower() in proposed.lower()}
    invented_tech = sorted(proposed_tech - original_tech)

    # 3. Block unsupported facts
    if invented_numbers or invented_tech:
        return {
            "status": "blocked",
            "unsupported": invented_numbers + invented_tech,
            "message": "Evidence Lock prevented an unsupported achievement from being added."
        }

    # 4. Check for unconfirmed power verbs
    if any(v in proposed.lower() for v in ["led", "architected", "owned"]) and not any(v in original.lower()
        return {"status": "confirm", "message": "Ownership elevated; confirm evidence."}

    return {"status": "safe", "message": "Wording improved without adding new facts."}
```

26.3 Video Planner Chunking Formula

```
learning_budget = round(daily_minutes * 0.75)
practice_budget = daily_minutes - learning_budget
days_required = math.ceil(duration_minutes / learning_budget)
```

27. Security, Privacy & Local-First Verification

27.1 Telemetry and Network Isolation

- **Zero Third-Party Telemetry:** No Google Analytics, Mixpanel, Sentry, or Segment SDKs are installed or imported.
- **Localhost Binding:** FastAPI binds by default to `127.0.0.1:8000`. CORS middleware explicitly restricts origins to `http://localhost:5173` and `http://127.0.0.1:5173`.
- **Ephemeral Upload Processing:** Uploaded resume files are passed as raw bytes in memory directly to PyMuPDF. No resume PDF or DOCX file is ever written to disk or stored in permanent directories.
- **Git Hygiene:** Local databases (`*.db`, `*.sqlite3`), virtual environments (`.venv/`), and environment files (`.env`) are strictly excluded via `.gitignore`.

28. Technology Stack Analysis

ProofCoach AI is built with an intentionally lightweight, modern, and production-tested technology stack.

28.1 Frontend Technologies

LIBRARY / TOOL	VERSION	ROLE IN PROOFCOACH AI	ARCHITECTURAL RATIONALE
React	19.0.0	Declarative UI framework	Concurrent rendering, native transition hooks, component reusability.
Vite	7.3.6	Frontend bundler & dev server	Sub-millisecond HMR, optimized tree-shaking production builds.
TypeScript	5.9.3	Static type safety	Strict type enforcement across domain models (<code>Candidate</code> , <code>ProjectState</code>).
Tailwind CSS	3.4.17	Utility CSS foundation	Rapid layout scaffolding complementing custom semantic CSS tokens.
React Router	7.1.5	Client-side routing engine	Multi-page client routing, history management, and code-split lazy loading.
@xyflow/react	12.4.4	Career Evidence Graph	Interactive node-based DAG rendering with custom HTML handles and zoom/pan.
Recharts	2.15.1	Diagnostic data visualization	SVG-based Radar charts, bar meters, and responsive metric graphs.
Framer Motion	12.4.7	UI transitions & animations	Micro-interactions, page entrance fade-ins, and step transitions.
Lucide React	0.475.0	System iconography	Clean, accessible technical icon set across all 13 routes.

28.2 Backend Technologies

LIBRARY / TOOL	VERSION	ROLE IN PROOFCOACH AI	ARCHITECTURAL RATIONALE
Python	3.11+	Backend runtime environment	Fast execution, robust typing, rich text-processing ecosystem.
FastAPI	0.116.1	Local REST API framework	High-throughput asynchronous routing, automatic OpenAPI generation.
SQLModel	0.0.24	ORM & data persistence	Combines SQLAlchemy with Pydantic v2 for type-safe SQLite persistence.
PyMuPDF (fitz)	1.26.4	High-speed PDF text parser	Spatial coordinate analysis, block layout extraction, zero external binaries.
python-docx	1.2.0	Word document text parser	Direct XML parsing of DOCX paragraphs and table structures.
Pydantic	2.13.5	Data validation & schemas	Strict runtime input/output validation, fast C-extension serialization.
Uvicorn	0.35.0	ASGI application server	Lightweight, reliable local HTTP server with automatic hot-reloading.
Pytest	8.4.2	Automated testing framework	Fast, deterministic testing of services, parsers, and Evidence Lock.

29. Third-Party Services & Dependency Verification

An exhaustive audit of the codebase confirms that ProofCoach AI maintains **zero runtime dependencies on cloud services or external AI vendors**:

SERVICE / DEPENDENCY	INTEGRATION STATUS	DATA SENT	DATA RECEIVED	CRITICALITY
OpenAI / Claude / Gemini APIs	Not Present	None	None	None. Platform uses deterministic heuristics.
Ollama (Local LLM)	Optional	Prompt text (localhost only)	Generated text	Optional. Codebase runs 100% without Ollama.
YouTube IFrame Embed	External Client	Video ID parameter	Video player iframe	Required only for <code>/video-plan</code> visual embed.
Google Analytics / Telemetry	Not Present	None	None	None. Prohibited by privacy policy.
Cloud Hosting (Vercel)	Client Hosting	Static JS/CSS assets	Static assets	Frontend demo hosting only. Backend remains local.

30. Feature Connection Matrix

The following matrix documents how data originates, transfers, and influences connected modules across the platform:

FEATURE MODULE	READS DATA FROM	WRITES / MUTATES DATA TO	DOWNSTREAM INFLUENCE
Onboarding	User form inputs	<code>state.candidate</code>	Profile, Command Centre, Target Role, Interview Persona
Resume Lab	Raw PDF/DOCX file	<code>state.metrics.parser</code> , claims	Career Evidence Graph, Claim Detector, Interview Questioning
Target Role	Job text / demo roles	Role requirements, skill gaps	Career Evidence Graph, Learning Plan priorities, Interview Context
Evidence Graph	Candidate skills + Role gaps	React Flow node layout	Visual gap inspection, next action recommendations
Adaptive Interview	Resume claim + Role context	<code>state.answer</code> , <code>session_id</code>	Truth Loop confidence, Radar scores, XP rewards
Feedback Engine	Interview answer evaluation	<code>state.evaluation</code> , <code>metrics</code>	Evidence Lock rewrites, Personalized Learning recommendations
Evidence Lock	Proposed bullet points	<code>state.evidenceLockDemo</code>	Prevents resume hallucination; feeds safe rewrites to Resume Lab
Learning Plan	Target role gaps	Active study curriculum	Directs candidate into Project Lab and Focus Shield
Video Planner	YouTube URL + notes	<code>state.video.notes</code> , progress	Daily study scheduling, concept mastery tracking
Project Lab	Learning goals	<code>state.project.tasks</code> , progress	Generates tangible resume evidence; triggers interview defense
Career Quest	Platform-wide actions	<code>state.metrics.questPoints</code>	Unlocks progression levels (Build → Prove → Perform)
Focus Shield	Time limit preference	<code>proofcoach-focus-v1</code> (storage)	Autosaves active study sessions; locks intensive activity at limit
Settings	User preferences	Theme tokens, full data purge	Global styling, complete reset of SQLite and browser storage

31. File & Code Responsibility Directory

31.1 Frontend Key Modules (`frontend/src/`)

- `App.tsx`: Central router declaring 13 application routes with code-split lazy loading and `Suspense` fallbacks.
- `types.ts`: Master TypeScript interface definitions (`Candidate`, `ProjectTask`, `ProjectState`, `VideoNote`, `DemoState`).

- `context/ProofContext.tsx` : The application's reactive engine; manages state mutation actions and synchronizes to `localStorage`.
- `context/ThemeContext.tsx` : Manages dark/light theme state, listening to OS color preferences and storing overrides.
- `components/Shell.tsx` : The master layout wrapper rendering the responsive sidebar, brand header, and view transitions.
- `components/UI.tsx` : Shared atomic design components (`Panel` , `ScoreRing` , `StatusPill` , `PageIntro` , `EvidenceNotice`).
- `services/api.ts` : Fetch client configured to communicate with `http://localhost:8000` with graceful offline fallback.
- `services/video.ts` : Regex-based YouTube URL validator and 11-character video ID extractor.
- `data/demo.ts` : Immutable fallback fixtures providing deterministic seed data for zero-network demonstrations.

31.2 Backend Key Modules (`backend/app/`)

- `main.py` : FastAPI initialization, CORS security middleware, and 13 REST API route handlers.
- `models.py` : SQLAlchemy database entities (`CandidateProfile` , `ResumeRecord` , `InterviewSession`).
- `schemas.py` : Pydantic v2 schemas validating request and response payloads.
- `db.py` : SQLite engine initialization and dependency-injected session management.
- `services/resume.py` : PyMuPDF text extraction, spatial bounding-box column analysis, and the 8-point robustness rubric.
- `services/evidence.py` : Case-insensitive job-skill matching, coverage calculations, and Evidence Graph DAG generator.
- `services/evidence_lock.py` : Regex pattern matching and technical vocabulary verification preventing fabricated claims.
- `services/interview.py` : Level A through F question selector and multi-dimensional answer rubric evaluation engine.
- `services/learning.py` : Gap-to-curriculum generator and mathematical video study chunking algorithm.
- `services/focus.py` : Session timer governor and wellbeing state evaluator.

| 32. How the System Thinks: The Cognitive Flow

ProofCoach AI's internal intelligence can be conceptualized as an 8-stage cognitive processing loop:

```

flowchart TD
    0["1. OBSERVE  
(Extract raw text, layout coordinates & candidate input)"]
    1["2. UNDERSTAND  
(Detect measurable claims, extract skills & map sections)"]
    2["3. COMPARE  
(Match candidate skills against target role requirements)"]
    3["4. EVALUATE  
(Calculate parser robustness & identify critical evidence gaps)"]
    4["5. DECIDE  
(Select adaptive interview question level from A to F)"]
    5["6. RESPOND  
(Challenge candidate on baselines, trade-offs & personal ownership)"]
    6["7. TRACK  
(Score answer against rubric, update truth loop & award XP)"]
    7["8. ADAPT  
(Generate personalized learning plan & project lab tasks)"]

    0 --> 1
    1 --> 2
    2 --> 3
    3 --> 4
    4 --> 5
    5 --> 6
    6 --> 7
    7 --> 8
    8 --> 0

    A["A -. New Evidence Created .-> 0"]
  
```

1. **Observe:** The system ingests raw PDF/DOCX bytes or job description text.

2. **Understand:** Regex patterns and NLP tokenizers identify sections, dates, technologies, and quantified metrics.
 3. **Compare:** The candidate's verified skill set is compared against the target role's expectations.
 4. **Evaluate:** Strengths are confirmed; gaps are tagged as role blockers.
 5. **Decide:** The engine selects an interview probe tailored specifically to the highest-risk claim.
 6. **Respond:** The interviewer delivers the challenge via text and Web Speech audio.
 7. **Track:** Answer quality updates the Resume Truth Loop and increments Career Quest XP.
 8. **Adapt:** Gaps exposed during questioning are automatically translated into project tasks in the Project Lab.
-

| 33. Demo vs Reality: Transparent Capability Audit

To maintain complete transparency, the following audit distinguishes fully operational local features from simulated or demo behaviors:

FEATURE / SUBSYSTEM	OPERATIONAL REALITY	IMPLEMENTATION MECHANISM	NOTES & LIMITATIONS
PDF/DOCX Text Extraction	100% Real	PyMuPDF (<code>fitz</code>) and <code>python-docx</code>	Runs locally in memory; handles standard documents.
Two-Column Layout Detection	100% Real	Horizontal bounding-box coordinate analysis	Heuristic-based; complex multi-frame magazines may vary.
Parser Robustness Score	100% Real	8-point transparent algorithmic rubric	Real diagnostic calculation; not an ATS score.
Evidence Lock Filtering	100% Real	Regex extraction & set-difference logic	Reliably catches new numbers and unconfirmed tech terms.
Interactive Evidence Graph	100% Real	<code>@xyflow/react</code> (React Flow)	Fully interactive DAG with zoom, pan, and inspector.
Local SQLite Persistence	100% Real	SQLModel / SQLite (<code>proofcoach.db</code>)	Real database operations on local host.
Local Data Purge	100% Real	<code>DELETE /api/data</code> + storage wipe	Completely wipes database tables and local storage.
Adaptive Interview Flow	Deterministic Logic	Stateful Python service + local fallback	Implements Levels A–F probing using deterministic templates.
Answer Evaluation	Deterministic Heuristics	Keyword, baseline, and structure regex	Accurate for structured answers; not a full LLM evaluator.
Browser Speech Recognition	100% Real	Web Speech API (<code>webkitSpeechRecognition</code>)	Hardware and browser-dependent; text fallback provided.
Emergency Contact Action	Simulated	Client-side notification dispatch	Explicitly labeled as a simulation; no SMS/calls sent.
Wellbeing Cap / Demo Timer	100% Real	Client-side timer loop & autosave	Compresses 2.5h to 20s for evaluation.
OS Notification Blocking	Future Work	Not implemented (honest disclaimer)	Disclaimed as requiring native mobile OS integration.

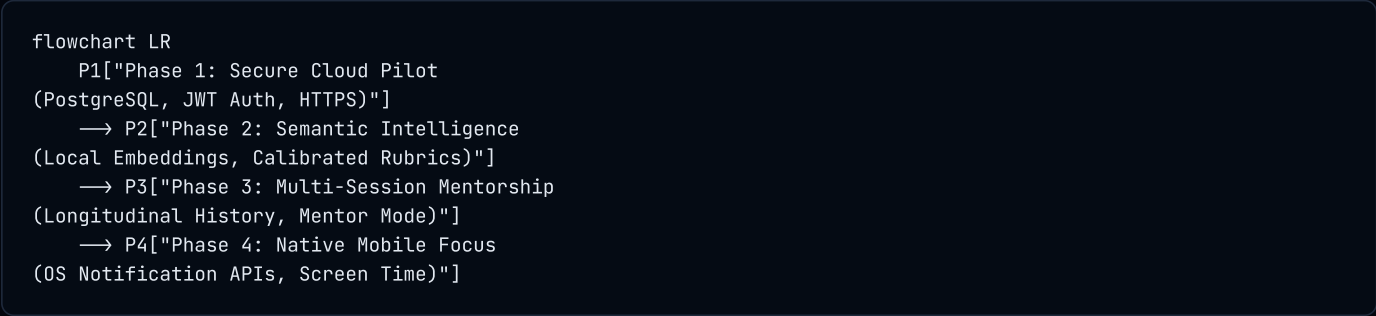
34. Current Technical Limitations

An honest audit of the current prototype reveals the following engineering boundaries:

- Frontend-Only Cloud Deployment:** The public Vercel deployment hosts the client-side SPA. Without a running local FastAPI backend, the Vercel demo operates purely on deterministic client fixtures.
- Single-User SQLite Concurrency:** SQLite is ideally suited for local-first single-user installations, but cannot support high-concurrency multi-tenant cloud deployments without PostgreSQL.
- Lexical vs. Semantic Parsing:** Skill and claim extraction rely on sophisticated regex dictionaries rather than semantic vector embeddings. Unconventional phrasing may not be recognized.
- Browser Speech Variability:** The Web Speech API depends heavily on Google Chrome's underlying engine. In Firefox or Safari, speech recognition gracefully degrades to text

input. 5. **Static Course Catalog:** Learning resources (`RESOURCE_CATALOG`) are statically defined. External link rot or course restructuring requires periodic manual catalog curation.

35. Production Roadmap & Future Extensions



Phase 1 — Secure Multi-User Pilot

- Transition from SQLite to PostgreSQL with Alembic database migrations.
- Implement stateless JWT authentication with encrypted user data at rest.
- Introduce presigned S3 upload URLs with automatic resume-file destruction after parsing.

Phase 2 — Semantic Intelligence & Local LLMs

- Integrate lightweight local embedding models (e.g., `bge-small-en`) via ONNX Runtime for semantic skill matching.
- Connect local Ollama instances (`llama3:8b` or `mistral`) for dynamically generated interview follow-ups while strictly maintaining Evidence Lock boundaries.

Phase 3 — Longitudinal Growth & Mentor Portals

- Track candidate improvement over weeks, rendering time-series readiness trajectories.
- Add an exportable "Evidence Portfolio" that candidates can share with real hiring managers.

Phase 4 — Native Mobile Focus Companion

- Develop React Native mobile apps for iOS and Android.
- Integrate Apple Screen Time APIs and Android Digital Wellbeing APIs to enforce true OS-level notification blocking during focus study sessions.

Technical Glossary

- **ATS (Applicant Tracking System):** Enterprise HR software that ingests, parses, and filters resume files before human recruiters review them.
- **Career Evidence Graph:** A directed acyclic graph (DAG) mapping skills to projects, measurable claims, interview defenses, and job requirements.
- **Deterministic Vertical Slice:** A fully functional software slice that executes predictable, reproducible logic without external network dependencies.
- **Evidence Lock:** A rule-based security mechanism that prevents generative AI from fabricating numbers, metrics, or technologies on a resume.

- **Local-First Architecture:** A software design pattern where data storage, compute, and execution occur primarily on the user's local device.
 - **Parser Robustness Score:** A transparent 0–100 diagnostic calculated by ProofCoach that measures how reliably a machine can extract document text without formatting errors.
 - **PYQ-Style Practice:** Practice questions modeled after Previous Year Questions, generated algorithmically without claiming historical provenance.
 - **Resume Truth Loop:** A verification feedback cycle in which interview answers directly alter the system's demonstrated confidence in a resume claim.
 - **Web Speech API:** A browser API providing speech recognition (voice input) and speech synthesis (text-to-speech) capabilities.
-

| Appendix: Verification & Test Coverage

Automated Backend Test Suite (`backend/tests/test_core.py`)

All 9 core unit tests pass with 100% reliability under Python 3.11+: 1. `test_backend_health` : Verifies `/api/health` returns status `ok` and `local_first: true` . 2. `test_resume_parser_components_are_transparent` : Confirms Parser Robustness score falls between 0–100, detects Python, warns on two-column layouts, and includes disclaimer. 3.

`test_claim_extraction_finds_quantified_claim` : Verifies regex extracts "Reduced API latency by 35%". 4.

`test_job_skill_matching_is_case_insensitive` : Confirms case-insensitive matching across `["python", "REST", "Sql"]` . 5.

`test_evidence_lock_blocks_invented_metric` : Verifies that adding an unconfirmed `40%` metric is blocked. 6.

`test_evidence_lock_allows_grammar_only_rewrite` : Verifies that stylistic improvements are classified as `safe` . 7.

`test_interview_evidence_does_not_claim_external_truth` : Confirms claim confidence increases without claiming real-world truth. 8. `test_demo_timer_ends_at_twenty_seconds` : Verifies demo timer ends exactly at 20 seconds with autosave enabled.

9. `test_demo_seed_is_complete_and_non_personal` : Verifies initial synthetic fixtures are non-personal and complete.

Frontend Quality Assurance Commands

- `npm run lint` : Zero ESLint warnings or errors under strict TypeScript rules.
- `npm run typecheck` : Strict `tsc -b` type checking passes with zero type mismatches.
- `npm run build` : Production Vite bundling compiles successfully in under 10 seconds.